GOVERNMENT OF KARNATAKA

DEPARTMENT OF SCHOOL EDUCATION (PRE-UNIVERSITY)

REVISED QUESTION BANK (2024 – 25)

SUBJECT:

BASIC MATHEMATICS (75)

SECOND YEAR PUC

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The revised Question Bank for 2024–25 has been prepared in adherence to the syllabus prescribed by the Pre-University Board of Karnataka. The objective of this Question Bank is to serve as a comprehensive tool that enhances students' understanding and preparation for examinations. A great deal of effort has been put into making this question bank a valuable resource for students.

To assist students, questions are organized chapter-wise. They are divided into one-mark, two-mark, three-mark, four-mark and five-/six-mark categories. Under the one-mark section, both multiple-choice questions and very short answer types (VSA) are included, in compliant with the latest blueprint.

By providing a wide range of questions and structured content, this question bank aims to facilitate effective study habits and reinforce key concepts. We hope that students will leverage this resource to boost their confidence and achieve academic success. Students are encouraged to utilize this question bank to the fullest.

Wishing all students the best in their studies!

The total weightage from each chapter is given below:

Sl.No	Chapter Name	Total Marks
1.	Matrices and Determinants	13
2.	Permutations and Combinations	8
3.	Probability	3
4.	Binomial Theorem	4
5.	Partial Fractions	5
6.	Mathematical Logic	6
7.	Ratios and Proportions	8
8.	Bill Discounting	5
9.	Stocks and Shares	3
10.	Learning Curve	5
11.	Linear Programming Problem	5
12.	Sales Tax and VAT	3
13.	Heights and Distances	4
14.	Compound Angles, Multiple and Submultiple Angles and Transformation Formulae	7
15.	Circles	6
16.	Parabola	4
17.	Limits ans Continuity	7
18.	Differential Calculus	8
19.	Application of Derivatives	5
20.	Indefinite Integrals	5
21.	Definite Integrals and Application to Areas	6
	Total marks	120

Chapter 1: MATRICES AND DETERMINANTS

One Mark Questions:

1. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ then adj A is

a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ b) $\begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$ c) $\begin{bmatrix} -4 & -2 \\ -3 & 1 \end{bmatrix}$ d) $\begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}$

2. If $A = \begin{bmatrix} 1 & 2 & 4 \\ -1 & 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -4 & -1 \\ 1 & 5 & -2 \end{bmatrix}$ then (A+B) is

a) $\begin{bmatrix} 4 & 2 & -3 \\ 0 & 8 & 4 \end{bmatrix}$ b) $\begin{bmatrix} 4 & -2 & 3 \\ 0 & 8 & -4 \end{bmatrix}$ c) $\begin{bmatrix} -4 & 2 & 3 \\ 0 & -8 & 4 \end{bmatrix}$ d) $\begin{bmatrix} 4 & -2 & -3 \\ 0 & -8 & -4 \end{bmatrix}$

3. The value of $\begin{vmatrix} 400 & 404 \\ 408 & 412 \end{vmatrix}$ is

a) -23 b) 23 c) 32 d) -32

4. If $A = \begin{bmatrix} 1 & -3 & 5 \\ 6 & 2 & 4 \end{bmatrix}$ then $5A'$ will be

a) $\begin{bmatrix} 6 & 2 & 4 \\ 1 & -3 & 5 \end{bmatrix}$ b) $\begin{bmatrix} 5 & 30 \\ -15 & 10 \\ 25 & 20 \end{bmatrix}$ c) $\begin{bmatrix} 5 & -15 & 25 \\ 30 & 10 & 20 \end{bmatrix}$ d) $\begin{bmatrix} 6 & 1 \\ 2 & -3 \\ 4 & 5 \end{bmatrix}$

5. If $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ then A' is

a) $\begin{bmatrix} -2 & -3 & 1 \\ -1 & 1 & 0 \end{bmatrix}$ b) $\begin{bmatrix} 2 & 1 \\ 3 & -1 \\ -1 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 2 & 3 & -1 \\ -1 & 1 & 0 \end{bmatrix}$ d) $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & -1 \end{bmatrix}$

6. If $A = \begin{bmatrix} 1 & -3 \\ 2 & 4 \end{bmatrix}$ then the matrix 2A is

a) $\begin{bmatrix} 2 & -6 \\ 4 & 8 \end{bmatrix}$ b) $\begin{bmatrix} 2 & 4 \\ -6 & 8 \end{bmatrix}$ c) $\begin{bmatrix} 8 & -6 \\ 4 & 2 \end{bmatrix}$ d) $\begin{bmatrix} 2 & 4 \\ 1 & -3 \end{bmatrix}$

7. If $\begin{vmatrix} 3 & x \\ 4 & 5 \end{vmatrix} = 0$ then the value of x is

a) $\frac{15}{4}$ b) 9 c) $\frac{4}{13}$ d) 13

8. If $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ then the matrix 2A is

a) $\begin{bmatrix} 2 & -4 \\ 6 & 8 \end{bmatrix}$ b) $\begin{bmatrix} 2 & 4 \\ -6 & 8 \end{bmatrix}$ c) $\begin{bmatrix} 2 & -6 \\ 4 & 2 \end{bmatrix}$ d) $\begin{bmatrix} 2 & 4 \\ 1 & -3 \end{bmatrix}$

9. The Value of $\begin{vmatrix} 3200 & 3201 \\ 3202 & 3203 \end{vmatrix}$ is

a) 4 b) 0 c) -2 d) 2

10. The value of $\begin{vmatrix} 2011 & 2012 \\ 2013 & 2014 \end{vmatrix}$ is

a) 2 b) 0 c) -2 d) 1

11. If $\begin{vmatrix} x & 3 \\ 12 & x \end{vmatrix} = 0$ then the value of x is

a) ± 12 b) 12 c) ± 6 d) 6

12. If $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ then the matrix 3A is

a) $\begin{bmatrix} 3 & -6 \\ 6 & 8 \end{bmatrix}$ b) $\begin{bmatrix} 3 & 4 \\ -6 & 8 \end{bmatrix}$ c) $\begin{bmatrix} 3 & -6 \\ 9 & 12 \end{bmatrix}$ d) $\begin{bmatrix} 3 & 4 \\ 1 & -3 \end{bmatrix}$

13. If $A = \begin{bmatrix} 2 & -x \\ x & -7 \end{bmatrix}$. Find $A + A'$.

a) $\begin{bmatrix} 2 & -6 \\ 6 & -7 \end{bmatrix}$ b) $\begin{bmatrix} 2 & 4 \\ -6 & 7 \end{bmatrix}$ c) $\begin{bmatrix} 4 & -6 \\ 9 & -14 \end{bmatrix}$ d) $\begin{bmatrix} 4 & 0 \\ 0 & -14 \end{bmatrix}$

14. If $A = \begin{bmatrix} 1 & 2 & 4 \\ -1 & 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -4 & -1 \\ 1 & 5 & -2 \end{bmatrix}$ then A-B is

a) $\begin{bmatrix} -2 & 6 & 5 \\ -2 & -2 & 0 \end{bmatrix}$ b) $\begin{bmatrix} 2 & -2 & 3 \\ 0 & 8 & -4 \end{bmatrix}$ c) $\begin{bmatrix} -4 & 2 & 3 \\ 0 & -8 & 4 \end{bmatrix}$ d) $\begin{bmatrix} 4 & -2 & -3 \\ 0 & -8 & -4 \end{bmatrix}$

15. If $A = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & 5 \end{bmatrix}$ then AB is

a) $\begin{bmatrix} -12 & 4 & 20 \\ -9 & -2 & 0 \end{bmatrix}$ b) $\begin{bmatrix} 12 & -2 & 3 \\ 9 & 8 & -4 \end{bmatrix}$ c) $\begin{bmatrix} 1 & 2 & 3 \\ -9 & 3 & 4 \end{bmatrix}$ d) $\begin{bmatrix} 12 & 4 & 20 \\ -9 & -3 & -15 \end{bmatrix}$

16. If $\begin{bmatrix} 2 & x & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \end{bmatrix}$ then x is

a) $\frac{1}{4}$ b) $\frac{-3}{4}$ c) $\frac{3}{4}$ d) $\frac{-1}{4}$

17. If $A = \begin{bmatrix} 3 & -1 \\ 2 & -4 \end{bmatrix}$ then A^2 is

a) $\begin{bmatrix} 7 & 1 \\ -2 & 14 \end{bmatrix}$ b) $\begin{bmatrix} 9 & 1 \\ 4 & 16 \end{bmatrix}$ c) $\begin{bmatrix} 9 & -1 \\ 4 & 16 \end{bmatrix}$ d) $\begin{bmatrix} 7 & -1 \\ 2 & -14 \end{bmatrix}$

18. If $A = \begin{bmatrix} -4 & -3 \\ -2 & -1 \end{bmatrix}$ then adj A is

a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ b) $\begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$ c) $\begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$ d) $\begin{bmatrix} -1 & 3 \\ 2 & -4 \end{bmatrix}$

19.If $A = \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$ then A^2 is

a) $\begin{bmatrix} 1 & -4 \\ 0 & 9 \end{bmatrix}$ b) $\begin{bmatrix} 1 & 4 \\ 0 & -9 \end{bmatrix}$ c) $\begin{bmatrix} 1 & 2 \\ 4 & 16 \end{bmatrix}$ d) $\begin{bmatrix} 1 & -1 \\ 2 & 6 \end{bmatrix}$

20.If $\begin{vmatrix} x & 4 \\ 4 & x \end{vmatrix} = 0$ then the value of x is

a) ± 4 b) 4 c) ± 2 d) 2

<u>Two Marks Questions:</u>

1. If $A = \begin{bmatrix} 1 & 3 & -1 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -1 & 2 \\ 1 & 3 & -2 \end{bmatrix}$ Find $A - 3B$.
2. If $A = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 & 2 \end{bmatrix}$. Find AB.
3. If $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & -2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & 4 \\ 1 & 3 & 2 \end{bmatrix}$ Find $2A - 3B$.
4. If $\begin{bmatrix} 2 & 3 \\ 7 & 5 \end{bmatrix} + \begin{bmatrix} 2 & x-2 \\ y-1 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 7 & 10 \end{bmatrix}$. Find x and y.
5. If $\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 2 & x-3 \\ y-4 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 2 & 3 \end{bmatrix}$. Find x and y.
6. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$ Find $A'B$.
7. If $\begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$ Find x and y.
8. If $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ Find AA'.
9. If $A = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$. Prove that $A^2 - 4A - 5I = 0$, where I is unit matrix and 0 is the null matrix.
10. If $A = \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix}$. Prove that $A^2 - A - 3I = 0$
11. Solve for x: $\begin{bmatrix} x^2 & 3 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 5x & 1 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ 5 & 5 \end{bmatrix}$
12. If $B + A = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $B - A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix}$. Find B
13. If $A = \begin{bmatrix} 3 & -2 & 4 \\ 2 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 & 4 \\ 6 & 2 & 5 \end{bmatrix}$ verify that $(A - B)' = A' - B'$.

14.If $A = \begin{bmatrix} 2 & -2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ -2 & 0 \end{bmatrix}$. Find AB.

15.If $X = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 4 \end{bmatrix}, Y = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$ $and\ Z = \begin{bmatrix} 2 & 1 \end{bmatrix}$. Verify X(YZ)=(XY)Z.

<u>Three Marks Questions:</u>

1. Solve by Cramer's rule: 2x+3y=11; x-y=3.
2. Solve by Cramer's rule: 3x+4y=7; 7x-y=6.
3. Solve by Cramer's rule: 3x+2y=8; 4x-3y=5.
4. Solve by Cramer's rule: 2x+y=1; x-3y=4.
5. Solve by Cramer's rule: 5x-7y=3; 7x-5y=9.
6. If$A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ 2 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ -2 & 4 \end{bmatrix}$ Verify that A(BC)=(AB)C.
7. If $A = \begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 \\ 6 & 2 \end{bmatrix}$. Show that $(AB)' = B'A'$
8. If A+B = $\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$ and 2A-B = $\begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix}$. Find A and B.
9. Find matrix A and B if $A + B = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$ and $A - B = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

10.If $A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ 2 & 1 \end{bmatrix}$. Show that $(AB)' = B'A'$.

11.If$A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ 2 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ -2 & 4 \end{bmatrix}$ Verify that A(B+C)=AB+AC.

12.Find x if $\begin{bmatrix} x^3 & 1 \\ 3 & 3 \end{bmatrix} + \begin{bmatrix} -2x^2 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -x & 4 \\ 4 & 7 \end{bmatrix}$.

13.If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $A + 2B = A^2$, find B.

14.Find x and y if $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \end{bmatrix}$.

15.If 2A+B = $\begin{bmatrix} 3 & -1 \\ -2 & 5 \end{bmatrix}$ and A-2B = $\begin{bmatrix} 4 & 2 \\ -1 & 5 \end{bmatrix}$. Find A and B.

<u>Five Marks Questions:</u>

1. Solve by matrix method x+y+z = 5; 2x+y-z=2; 2x-y+z = 2.
2. Solve by matrix method x+y-z = 1; 3x+y-2z=3; x-y-z = -1.

3. Solve by matrix method 3x+y+2z = 3; 2x-3y-z=-3; x+2y+z = 4.
4. Solve by matrix method x-y+2z = 3; 2x+z=1; 3x+2y+z = 4.
5. Solve by matrix method x-y-2z = 3; 2x+y+z=5; 4x-y-2z = 1.

Chapter 2: PERMUTATIONS AND COMBINATIONS

Multiple Choice Questions

1. The number of ways 9 flowers of different can be strung together to form a garland is
   a) 9! b) 8! c) $\frac{9!}{2}$ d) $\frac{8!}{2}$
2. The number of ways in which 10 people ca be seated around a table is
   b) 9! b) 10! c) $\frac{9!}{2}$ d) $\frac{10!}{2}$
3. The number of straight lines and triangles that can be formed from 12 points in a plane o which 5 are collinear is
   a) 57 ,210 b) 57 , 211 c) 56 , 210 d) 56 , 211
4. The number of parallelograms that can be formed from a set of five parallel lines intersecting another set of 6 parallel lines is
   a) 30 b) 25 c) 150 d) 100
5. The number of diagonals in a decagon is
   a) 45 b) 35 c) 55 d)10
6. The number of ways in which 6 people be selected out of 12 so that two particular members are always excluded is
   a) ${}^{12}P_6$ b) ${}^{12}P_4$ c) ${}^{10}P_4$ d) ${}^{10}P_6$
7. ${}^{5}P_3 : {}^{5}P_2 =$
   a) 3:1 b) 1:3 c) 2:3 d) 3:2
8. The value of ${}^{8}P_3$ is
   a) 336 b) 56 c) 24 d) 8
9. If ${}^{5}P_r=60$ then r
   a) 4 b) 3 c) 2 d)5
10. $If\ {}^{n}C_4 = {}^{n}C_5$ find the value of n.
   a) 20 b) 9 c) 8 d) 10

<u>One Mark Questions:</u>

1. Find the value of ${}^{8}P_3$
2. In how many ways can 9 soldiers stand in a queue?

3. In how many ways can four people occupy 6 vacant chairs.
4. If ${}^{5}P_{r}=60$ find r
5. If ${}^{n}P_{3}=210$ Find n
6. If ${}^{n}P_{3}:{}^{n}P_{2}=3:1$ Find n
7. How many 3 letter words can be formed using only consonants but each only once.
8. Find the number of permutations of the letters of the word MONDAY
9. How many 4 digit numbers can be formed using the digits 1, 2, 4, 5, 7, 8, 9 no digit being repeated in any number.
10. Find the total number of ways in which 8 different coloured beads can be strung together to form a necklace.
11. In how many ways can 9 flowers of different colours be strung together to form a garland.
12. In how many ways can 10 different precious stones be set to form a necklace.
13. In how many ways can 10 people be seated around the table.
14. In how many ways a committee of5 can be chosen from 10 students.
15. How many triangles can be formed from 8 non collinear points.
16. How many straight lines can be formed from 10 points if no three of them are collinear?
17. Find the number of permutations of the letters of the word “MATHEMATICS”.
18. If ${}^{n}C_{4} = {}^{n}C_{5}$ find the value of n.
19. Find x if ${}^{99}C_{40}+{}^{99}C_{x}={}^{100}C_{59}$
20. If $C_{10} = {}^{n}C_{15}$ find the value of n.
21. Find x if ${}^{9}C_{x}+{}^{9}C_{7}={}^{10}C_{7}$
22. In how many ways can the letters of the word “HOPPER” be arranged?
23. In how many ways can 5 letters be posted in 5 letter boxes if each box has one letter?
24. If there are 4 routes to go from A to B and 3 routes to go from B to C, in how many ways can you go from A to C via B.
25. In how many ways can you go from A to B and return to A from a different route, if there are 10 ways of travelling from A to B.
26. How many 3 digit numbers can be formed using digits 1,2,5,6,8
27. How many different signals can be made by taking 3 different coloured flags at a time from 7 different coloured flags?

28. How many 6 digit numbers can be formed from 2,7,6,1,9,8

Two Marks Questions:

1. In how many ways can 7 students and 4 teachers be seated in a row so that no two teachers are together
2. A box contains 5 red 4 black and 3 white shirts. How many selections of 8 shirts can be made if the selection contains exactly 4 red 2 blackand 2 white shirts.
3. In how many ways can 6 people be chosen out of 10 people if one particular person is always included?
4. Find the number of triangles and straight lines that can be formed out of 20 points in which 8 are collinear.
5. How many words can be formed from the letters of the word FRIEND which begin with F and end with D
6. Find the number of diagonals in a decagon.
7. Find the number of diagonals of a polygon of 20 sides.
8. Find the number of parallelograms that can be formed from 6 parallel lines intersecting another set of 4 parallel lines.
9. If a convex polygon has 170 diagonals, find the number of sides of the polygon.
10. In how many ways can 7 people be seated for a photograph if two particular people always want to be together
11. In how many ways can 7 math books 8 statistics books and 4 accountancy books be arranged in a shelf if books of the same subject are always together?
12. In how many ways can 9 boys and 6 girls be arranged in a row if no two girls are together?
13. Find the number of words that can be formed using all the letters of the word ENGINEERING having all the 3 E's together.
14. How many 4 digit even numbers can be formed from the digits 1, 2, 3,4,5,6 no digit being repeated.
15. In how many ways 10 beads of different colours are strung into a necklace if the red green and yellow beads are always together.

16. Find the number of ways in which 8 men be arranged around a table so that two particular men may not be next to each other.
17. In how many ways 6 gentlemen and 4 ladies are seated round a table so that no two ladies are together.
18. In how many ways can 6 boys be arranged in a circle if 2 particular boys always want to be together?
19. If ${}^{n}P_{r} = 3024$ ${}^{n}C_{r} = 126$ find r.
20. If ${}^{n}P_{r} = 24\,{}^{n}C_{4}$ find n.
21. If $C_8 = {}^{n}C_{12}$ find the value of ${}^{n}C_{5}$.
22. A person has 9 friends .In how many ways can he invite one or more of them to dinner.
23. Find the number of parallelograms that can be formed from a set of 5 parallel lines intersecting another set of 6 parallel lines.
24. In how many ways can 6 people be selected out of 12 people so that a) two particular people must be included b) Two particular members must be excluded
25. In a party each person shakes hand with everyone else. If there are 25 members in the party, calculate the number of handshakes.
26. In how many ways can 3 boys and 4 girls be arranged in a row so that all the three boys are together?
27. In how many ways can 5 men and 4 women be seated in a row such that women occupy the even places?
28. In how many ways can 7 persons be seated in a row if two persons always occupy the end seats?
29. In how many ways can 6 red and 4 white marbles be chosen from a bag containing 10 red and 6 white marbles.
30. In how many ways can 6 people be chosen out of 10 people if one particular person is always included?
31. In how many ways can 7 persons be seated in a row if two persons always occupy the even seats?

32. In how many ways the word “CARROM” be arranged such that the 2R’s are always together.
33. How many words can be formed from the letters of the word “FRIEND” which begin with F and end with D.?
34. How many four digit numbers can be formed with the 0, 2,3,5,7 such that no digits are repeated.
35. How many three digit numbers ending with 7 can be formed using the digits 1, 2,5,7,8 digits cannot be repeated.
36. How many 6 digit numbers can be formed from the digits 1, 2, 3,4,5,6 (no digit being repeated) which are divisible by 5.
37. In how many ways can 7 students and 4 teachers be seated in a row such that no two teachers are together?
38. In how many ways can 7 books on art, 5 books on fiction, 4 books on adventure be arranged so that books of the same category are always together.
39. Find the number of ways in which 4 boys and 5 girls can be arranged in a row so that no two boys are together and no two girls are together.
40. In how many ways 10 examination papers are arranged so that best and worst papers never come together.
41. In how many ways can the letters of the word “LAUGHTER” be arranged so that the vowels are always together?
42. In how many ways can the letters of the word “HEXAGON” be arranged so that the vowels are in even places

<u>Three Marks Questions:</u>

Note: The questions with multiple sub questions can be asked for two or three marks. For two mark questions only two sub questions will be asked and for three mark questions only 3 sub questions will be asked.

1. A family of 4 brothers and 3 sisters is to be arranged for a photograph in one row. In how many ways can they be seated if
   a) all the sisters sit together

b) no two sisters sit together

2. Find the number of ways in which 4 boys and 5 girls can be arranged in a row so that no two boys are together and no two girls are together.
3. In how many ways can 6 boys and 6 girls be arranged in a row so that

   a) All girls are together

   b) All boys are not together

   c) Boys and girls are arranged alternatively.

   d) Two particular girls are always together

   e) No two girls are together
4. In how many ways 7 math books 8 statistics books and 4 accountancy books can be arranged in a shelf if

   a) Maths books are always together

   b) Books of the same subject are always together.
5. Find the number of permutations of the letters of the word MISSISSIPPI, In how many of these

   a) The 4 S's are together

   b) The 4 S's are not together

   c) Begin with MISS

   d) Begin with SIP
6. Find the number of permutations of the letters of the word 'COMMITTEE'

   a) How many of them begin with T and end with T

   b) In how many all the vowels are together

   C) In how many no two vowels are together

   d) How many of them end with MITE.
7. Find the number of permutations of the letters of the word ASSASSINATION. In how many of these

   a) the vowels are in even places

   b) vowels are in odd places

   c) The word NATION is always present together.

   d) begin with 'AS' and end with 'AS'

29. In how many ways can 7 English books, 8 Kannada books and 5 Hindi books be arranged in a shelf? In how many of these

a) All Kannada books are together

b) No two English books are together

c) Books of the same language are together

d) All Kannada books are together and all Hindi books are together.

30. How many 4 digit numbers can be formed using digits 0, 1, 2, 3, 4, 5, 6 (No digit can be repeated. How many of them are

a) Even b) Odd c) divisible by 5

31. A man has 10 relatives, 4 of them are ladies 3 gentlemen and 3 children. In how many ways can he invite 7 relatives to a dinner party so that

a. There are exactly 2 ladies, 3 gentlemen and 2 children

b. There are exactly 2 gentlemen and at least 3 ladies.

c. There are exactly 3 children, at least 1 lady and at least two gentlemen

32. From a class of 12 boys and 10 girls, 10 students are to be chosen for a competition, including at least 4 boys and at least 4 girls. The 2 girls who won prize last year should be included. In how many ways can the selection be made?

33. A box contains 5 red, 4 black and 3 white balls. How many ways selection of 8 balls can be done if the selection contains

a. Exactly 4 red, 2 black and 2 white balls.

b. Atleast 3 red, atleast 3 black and atleast 1 white balls.

34. An examination paper consists of 12 questions divided into part A and B contains 7 questions in part A and part B contains 5 questions. A candidate is required to answer 8 questions selecting atleast 3 from each part. In how many ways can the candidate select the questions?

35. A team of eleven is to be chosen out of 16 cricket players of whom 4 are bowlers and 2 wicket keepers. In how many ways can the team be chosen so that

a) There atleast 3 bowlers and 1 wicket keeper.

b) There atleast 3 bowlers and atleast one wicket keeper.

36. Out of 4 officers and 10 clerks in an office a committee consisting of 2 officers and 3 clerks is to be formed. In how many ways can this be done if

a) any officer and any clerk can be included

b) one particular clerk must be on committee

c) one particular officer can be on the committee.

37. A committee of 5 are to be formed from 8 Americans and 5 Anglo- Indians. In how many ways can this be done when the committee contains

a) exactly two Anglo-Indians

b) atleast two Anglo-Indians.

38. A candidate is required to answer 6 out of 12 questions which are divided into 2 groups containing 6 questions in each group. Find the number of choices he has if he cannot answer more than five questions from any group.

39. Find the number of permutations of the letters of the word "ENGINEERING" How many of these

a) Begin with E and end with E

b) Have all the 3 E's together

c) Begin with GIN and end with GRIN

d) All the vowels are not together

e) No two vowels are together

Chapter 3: PROBABILITY

<u>Multiple Choice Questions:</u>

1. If E is an event of sample space S then $P(EUE^1) =$

a) $P(E)$ b) $P(E^1)$ c) $P(S)$ d) 0

2. If $P(A) = \frac{1}{2}$, $P(B) = 0$, then $P(A|B)$ is

a) 0 b) $\frac{1}{2}$ c) not defined d) 1

3. The probability of obtaining an even prime number on each die, when a pair of dice is rolled is

a) 0 b) $\frac{1}{3}$ c) $\frac{1}{12}$ d) $\frac{1}{36}$

4. If A and B are events such that $P(A|B) = P(B|A)$, then

a) $A \subset B$ but $A \neq B$ b) $A = B$ c) $A \cap B = \emptyset$ d) $P(A) = P(B)$

5. Two events A and B will be independent, if

a) A and B are mutually exclusive b) $P(A'B') = [1 - P(A)][1 - P(B)]$

c) $P(A) = P(B)$ d) $P(A) + P(B) = 1$

6. Let E and F be events of a sample space S of an experiment, then $P(E'|F) + P(E|F)$ is

a) $P(F)$ b) $P(E)$ c) $P(S)$ d) $P(E \cap F)$

7. If the events A and B independent then $P(A \cap B) =$

a) $P(A) + P(B)$ b) $P(A) = P(B)$ c) $P(A) \times P(B)$ d) $\frac{P(A)}{P(B)}$

8. If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$ then $P(A|B)$ is

a) $\frac{4}{13}$ b) $\frac{4}{9}$ c) $\frac{9}{13}$ d) $\frac{4}{7}$

9. If $P(A) = \frac{8}{15}$, $P(B) = \frac{7}{15}$ and $P(A \cap B) = \frac{2}{15}$ then $P(B|A)$ is

a) $\frac{2}{7}$ b) $\frac{7}{8}$ c) $\frac{1}{4}$ d) $\frac{2}{15}$

10. If $P(B) = 0.5$ and $P(A \cap B) = 0.32$ then $P(A|B)$ is

a) $\frac{8}{25}$ b) $\frac{1}{2}$ c) $\frac{4}{25}$ d) $\frac{16}{25}$

11. If $P(A) = 0.5$ and $P(B) = 0$ then find $P(A|B)$ is

a) $\frac{1}{2}$ b) 0 c) 1 d) not exists

12. If $P(A) = 0.4$, $P(B) = 0.8$ and $P(B|A) = 0.6$ then $P(A \cap B)$ is

a) 0.24 b) 0.3 c) 0.48 d) 0.96

13. If $P(A) = 0.3$ and $P(B) = 0.4$, A and B are mutually exclusive events then $P(A|B)$ is

a) 0 b) 1 c) -1 d) None of these

14. If $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{5}$, A and B are independent events then $P(A \cap B)$ is

a) $\frac{3}{25}$ b) $\frac{1}{5}$ c) $\frac{4}{5}$ d) $\frac{1}{25}$

15. If $P(E) = 0.6$, $P(F) = 0.3$ and $P(E \cap F) = 0.2$ then $P(E|F)$ is

$a)\ \frac{1}{3}$ b) $\frac{2}{3}$ c) 0 d) 3

16.If $A \cap B = \emptyset$ then $P(B|A)$ is

$a)\ \frac{1}{2}$ b) 1 c) 0 d) not exists

17.The chances of getting a doublet with two dice is

a) $\frac{3}{2}$ b) $\frac{1}{6}$ c) $\frac{5}{6}$ d) $\frac{5}{36}$

18.If $P(A) = 0.8$, $P(B) = 0.5$, $P(B|A) = 0.4$ then $P(A|B)$ is

a) $\frac{16}{25}$ b) $\frac{4}{5}$ c) $\frac{1}{2}$ d) $\frac{4}{25}$

19.If $P(A) = \frac{6}{11}$, $P(B) = \frac{5}{11}$ and $P(A \cup B) = \frac{7}{11}$ then $P(B|A)$ is

a) $\frac{5}{6}$ b) $\frac{6}{7}$ c) $\frac{4}{5}$ d) $\frac{2}{3}$

20.If $P(A) = \frac{4}{5}$ and $P(A \cap B) = \frac{7}{10}$ then $P(B|A) =$ is

a) $\frac{1}{10}$ b) $\frac{1}{8}$ c) $\frac{7}{8}$ d) $\frac{17}{20}$

21.If $P(A \cap B) = \frac{7}{50}$, $P(B) = \frac{17}{20}$, then $P(A|B) =$ is

a) $\frac{14}{85}$ b) $\frac{20}{50}$ c) $\frac{7}{17}$ d) none of these

<u>One Mark Questions:</u>

1. What is the probability of the sample space.
2. What is the Range of the probability.
3. What is the probability of an Impossible event.
4. What is the probability of an Sure event.
5. Find the probability of getting a head in a throw of a coin.
6. An unbiased die is rolled. Find the probability of getting a multiple of 3.
7. If $P(A) = \frac{3}{5}$ then Find $P(A')$
8. A coin is tossed thrice. Write the sample space.
9. Ram hits the target 7 times out of 10 shots. Find his probability of missing the target.

10. What is the probability of an event A if odds in favour are $3:5$.

11. What is the probability of an event A if odds against are $2:7$.

12. A card is drawn from a pack of 52 playing cards. What is the probability that it is a queen.

13. What is the probability of getting two heads when a coin is tossed twice.

14. If $P(A) = \frac{3}{4}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{4}$, find $P(A/B)$

15. A bag contains 8 red and 4 green marbles. Find the probability that a marble selected a random is red.

16. A bag contains 6 white beads and 4 red beads. A bead is drawn at random. What is the probability that the bead drawn is white.

17. A box contains 5 red balls, 8 green balls and 10 pink balls. A ball is drawn at random from the box. What is the probability that the ball drawn is either red or green.

18. What is the probability of getting a multiple of 3 when a die is thrown.

19. What is the probability that the child born is a girl?

20. A number chosen from 1 to 50. What is the probability that the number is a multiple of 8.

21. If A and B are mutually exhaustive events, then what is $P(AUB)$

22. A die is thrown. What is the chance of getting a face with dots as multiple of 2.

23. If A and B are independent events, find $P(B/A)$

<u>Two Marks Questions:</u>

1. A box contains 8 red marbles, 6 green marbles and 10 pink marbles. One marble is drawn at random from the box. What is the probability that the marble drawn is either red or green?
2. If $P(A) = \frac{1}{2}. P(B) = \frac{1}{3}, P(A \cup B) = \frac{7}{12}$ then find $P(B/A)$.
3. Two coins are tossed simultaneously, find the probability of getting a) getting exactly two heads b) atleast one head.
4. If A and B are Mutually exclusive events with $P(A) = \frac{2}{5},\ P(B) = \frac{1}{7}$, then find $P(A \cup B)$.
5. If $P(\overline{A}) = 0.65,\ P(AUB) = 0.65$. Find $P(B)$, if A and B are Mutually exclusive events.

6. An unbiased die is rolled. Find the probability of (a) getting a multiple of 3. (b) getting a prime Number.
7. Two cards are drawn at random from a well – shuffled pack of 52 cards, what is the probability that either both are Queens or both are king cards.
8. A die is thrown twice, what is the probability that atleast one of the two numbers is 6.
9. The probability of occurrence if two events A and B are $\frac{1}{4}$ and $\frac{1}{2}$ respectively. The probability of their simultaneous occurrence is $\frac{7}{50}$. What is the probability that neither A nor B occurs?
10. Two coins are tossed simultaneously. What is the probability of getting (a) atmost 1 tail (b) atleast 1 tail.
11. Three fair coins are tossed simultaneously. Find the probability of getting atleast one head and atleast one tail.
12. Two dice are rolled simultaneously. Find the probability of getting a doublet of even numbers.
13. A box contains 5 defective and 15 non defective bulbs. Two bulbs are chosen at random. Find the probability that both the bulbs are non-defective.
14. A problem in a question paper is given to 3 students in a class to be solved. The probability of their solving the problem are $0.5, 0.7$ and 0.8 respectively. Find the probability that the problem will be solved.
15. What is the probability that a randomly chosen two digit positive integer is a multiple of 3.
16. Two cards are drawn from a pack of 52 cards, what is the probability that both are face cards.
17. Tickets are numbered from 1 to 18 are mixed up together and one ticket is drawn at random. What is the probability that the ticket has a number which is a multiple of 2 or 3.
18. If the letters of the word *'RAMLEELA'* are arranged in random. What is the probability that it begins with $REEL$.
19. A committee of 4 has to be selected from 9 boys and 6 girls. What is the probability that the committee contains 2 boys and 2 girls.

20. If A and B are two events with probability 0.4 and 0.8 correponding to A and $A \cup B$. Find $P(B)$, if A & B are mutually exclusive.
21. Three of the six vertices of a regular hexagon are chosen at random. What is the probability that the triangle formed with 3 vertices is equilateral.

Three Marks Questions:

1. A box contains 4 defective and 6 non defective bulbs. Find the probability that atleast 3 bulbs are defective when 4 bulbs are selected at random.
2. A natural number is chosen at random from among the first 300. What is the probability that the number so chosen is divisible by 3 or 5.
3. The probability that a MBA aspirant will join IIM is $\frac{2}{5}$and that he will join XLRI is $\frac{1}{3}$. Find the probability that a) he will join IIM or XLRI b) he will join neither IIM nor XLRI
4. A die is thrown twice and sum of the numbers appearing is observed to be 9. What is the conditional probability that the number 4 has appeared atleast once.
5. A die is rolled. If the outcome is an odd number. What is the probability that it is a number greater than 1.
6. A couple has two children. Find the probability that both are boys, if it is known that a) one of the children is a boy b) older child is a boy.
7. A die is rolled twice. If the sum of the numbers facing upwards is even find the probability that both are odd.
8. Seven persons are to be seated in a row. Find the probability that 2 particular persons sit next to each other.
9. There are 20 girls and 60 boys in a class. Half of the girls and half of the boys are first class students. A student is selected at random. What is the probability that the student is either a boy or a first class holder.
10. Among the members of a committee, there are 75% males and 25% females. The probability that a male member becomes the president is 0.25 and probability that a female member becomes the president is 0.4. Find the probability that the person selected at

random becomes the president. [Hint: Person selected is Male and becomes President OR is female and becomes President]

11. Two cards are drawn from a pack of playing cards, one after the other. Find the probability of getting a heart in first draw and diamond in the second draw if the cards are drawn. A) without replacement b) with replacement
12. If three cards are drawn at random from a pack of 52 cards, what is the probability that atleast two of them are kings.
13. Two dice are rolled together. Find the probability of getting
    a) a doublet
    b) sum of 2 numbers on the dice as 11
    c) sum of 2 numbers on the dice are more than 11.
14. 3 fair coins are tossed simultaneously. Find the probability of
    a) getting 1 head
    b) getting at the most one head
    c) getting at least two heads.
15. Two cards are drawn from a pack of playing cards, one after the other. Find the probability of getting a queen in first draw and second draw, if the cards are drawn.
    a) without replacement b) with replacement
16. A bag contains 10 gold and 8 silver coins. Two successive drawings of 4 coins are made. Find the probability that the first drawing will get 4 gold coins and second 4 silver coins, if the coins are replaced before the second trial.
17. One card is drawn from a pack of 52 playing cards. Find the probability that
    a) card is either black or jack
    b) card is red
    c) card is diamond
18. A die is thrown. If F is the event “the number appearing is even” and E is the event “the number appearing is multiple of 3”. Then find whether E and F are independent.
19. A box contains 8 white and 9 red balls. Two balls are taken at random from the box. Find the probability that both of them are red, if a) the two balls are taken together b) the balls

are taken one after the other without replacement c) the balls are taken one after the other with replacement.

Chapter 4: BINOMIAL THEOREM

Three Marks Questions:

1. Using Binomial Theorem, expand the following

   a) $(1-2x)^5$

   b) $\left(\frac{2}{x}-\frac{x}{2}\right)^5$

   c) $(2x-3)^6$

   d) $\left(\frac{x}{3}-\frac{1}{x}\right)^5$

   e) $\left(x+\frac{1}{x}\right)^6$

   f) $\left(x^2+\frac{3}{x}\right)^4$

Four Marks Questions:

1. Find the value of $(1.01)^5$ using Binomial Theorem up to 5 decimal places.
2. Find the value of $(99)^5$ using Binomial Theorem up to 5 decimal places.
3. Find the value of $(1.2)^5$ using Binomial Theorem up to 5 decimal places.
4. Find the value of $(0.99)^5$ using Binomial Theorem up to 5 decimal places.
5. Find the value of $(98)^4$ using Binomial Theorem up to 5 decimal places.
6. Find the value of $(0.99)^4$ using Binomial Theorem up to 5 decimal places.
7. Find the value of $(1.02)^6$ using Binomial Theorem up to 5 decimal places.

Five Marks Questions:

1. Find the middle terms in the expansion of $\left(2x^2-\frac{1}{\sqrt{x}}\right)^{11}$
2. Find the middle terms in the expansion of $\left(2x-\frac{1}{x}\right)^{17}$
3. Find the middle terms in the expansion of $\left(\frac{2x^2}{3}-\frac{3}{2x}\right)^{10}$

4. Find the middle terms in the expansion of $\left(x-\frac{1}{y}\right)^{10}$

5. Find the middle terms in the expansion of $\left(\frac{2a}{3}-\frac{3}{2a}\right)^{6}$

6. Find the middle terms in the expansion of $\left(\frac{x}{2}+\frac{3}{x^2}\right)^{19}$

7. Find the co efficient of x^{-11}in $\left(\sqrt{x}-\frac{2}{x}\right)^{17}$

8. Find the co efficient of x^{5}in $\left(x+\frac{1}{x^2}\right)^{17}$

9. Find the co efficient of x^{18}in $\left(x^2+\frac{3a}{x}\right)^{15}$

10. Find the co efficient of x^{-7}in $\left(\sqrt{x}-\frac{4}{x^3}\right)^{21}$

11. Find the co efficient of x^{8} in $\left(3x^2-\frac{1}{2x}\right)^{10}$

12. Find the constant term in the expansion of $\left(2x^2-\frac{3}{x^3}\right)^{25}$

13. Find the term independent of x in $\left(\sqrt{x}+\frac{1}{3x^2}\right)^{10}$

14. Find the term independent of x in $\left(x^2-\frac{2}{x^3}\right)^{5}$

15. Find the term independent of x in $\left(\sqrt{x}-\frac{2}{x^2}\right)^{20}$

16. Simplify $\left(2+\sqrt{3}\right)^{5}+\left(2-\sqrt{3}\right)^{5}$

17. Simplify $\left(3+\sqrt{2}\right)^{4}+\left(3-\sqrt{2}\right)^{4}$

18. Simplify $\left(\sqrt{2}+1\right)^{6}+\left(\sqrt{2}-1\right)^{6}$

Chapter 5: PARTIAL FRACTIONS

<u>Five Marks Questions:</u>

Resolve the following into partial fractions.

1. $\frac{x+3}{(x-1)(x+2)(x-2)}$

2. $\frac{x^2}{(x+1)(x+2)(x+3)}$

3. $\frac{x-1}{(x)(x+2)(x+4)}$

4. $\frac{x^2-10x+13}{(x+1)(x^2-5x+6)}$

5. $\frac{x+8}{(2x^2-x-1)(x+1)}$

6. $\frac{3x+5}{(x+2)^2(x-3)}$

7. $\frac{1}{x(x^2-9)}$

8. $\frac{2x+21}{(x+2)(x-1)^2}$

9. $\frac{x}{(x+1)(x^2+3x+2)}$

10. $\frac{x^2}{(x+1)^2(x-5)}$

11. $\frac{2x^2+16x+29}{(x+3)^2(x+4)}$

12. $\frac{9}{(x+2)^2(x+1)}$

13. $\frac{3x+4}{(x+1)^2(x-1)}$

14. $\frac{x-1}{(x+2)(x^2-x-6)}$

15. $\frac{1+3x+2x^2}{(1-2x)(1-x^2)}$

16. $\frac{x}{(1+2x)^2(1-3x)}$

17. $\frac{9x-27}{(x+1)(x-2)^2}$

18. $\frac{2x+1}{(x-1)(x+2)(3x-1)}$

Chapter 5 :MATHEMATICAL LOGIC

<u>One Mark Questions:</u>

1. Symbolize the propositions “2 + 5 = 6 or all integers are rationals”.
2. Symbolize the propositions “3x = 9 and x<7”.
3. Symbolize the propositions “If 2 numbers are equal then their squares are not equal”
4. Write the converse of “If x (x-2) = 0 then x=2”
5. Write the inverse of “If the questions are easy then students score better marks”.
6. Write the contrapositive of “ If $x \in (A \cup B)\ then\ x \in A\ or\ x \in B$.
7. Write the converse of “ If $x^2 = y^2\ then\ \ x = y$.
8. Negate: “4 is an even integer or 7 is a prime number”.
9. Negate: $\sim p \rightarrow q$.
10. Symbolize the propositions “If oxygen is a gas then gold is a compound”.

<u>Two Marks Questions:</u>

1. If the compound proposition $p \rightarrow (q \vee r)$ is false, then find the truth values of p,q and r.
2. If the compound proposition $p \rightarrow (\sim q \vee r)$ is false, then find the truth values of p,q and r.

3. If $(\sim p \vee q) \wedge \sim r$ is a false proposition then find the truth values of p,q and r.
4. If the truth value of the proposition p,q and r are F,T and F respectively,then find the truth values of the compound proposition $(\sim p \rightarrow q) \vee r$.
5. If the truth value of the proposition p,q and r areT,T and F respectively,then find the truth values of the compound proposition$(p \vee r) \wedge q$.
6. If the truth value of the proposition p,q and r are F,T and F respectively,then find the truth values of the compound proposition$p \rightarrow (q \rightarrow r)$.
7. Negate: $q \vee [\sim(p \wedge r)]$.
8. Negate: 14 is a divisor of 48 and 28 is not divisible by 82.
9. Determine whether the proposition $[\sim p \wedge (p \vee q)]$ is a Tautology or a Contradiction or neither.
10. Write the converse and inverse of the implication " If the questions are easy then students score better marks".

<u>Three Marks Questions:</u>

1. Construct the truth table for $(p \vee q) \leftrightarrow (q \wedge p)$.
2. Construct the truth table for $\sim p \leftrightarrow (p \wedge \sim q)$.
3. Construct the truth table for $(p \wedge \sim q) \leftrightarrow \sim p$.
4. If p: $\sqrt[3]{3}$ is an integer, q: 2 is an odd number, r: 5 is a prime number then find the truth value of the proposition $(p \leftrightarrow q) \wedge r$.
5. Show that $\sim(p \wedge q) \equiv \sim p \vee \sim q$.
6. Show that $\sim(p \vee q) \equiv \sim p \wedge \sim q$.
7. Write the converse, inverse and contrapositive of "If x is less than 1 then it is a prime number".
8. Write the converse, inverse and contrapositive of " If oxygen is a gas then accountancy is easy or the child is brave".
9. Write the converse, inverse and contrapositive of "If $x^2 = y^2$ then "x = y".
10. Write the converse, inverse and contrapositive of " If $x \in (A \cup B)$ $then$ $x \in A$ or $x \in B$".

Five Marks Questions:

1. Show that $[\sim p \wedge (p \vee q)] \rightarrow q$ is a Tautology or a Contradiction.
2. Show that $[\sim(p \rightarrow \sim q)] \vee (\sim p \leftrightarrow q)$ is a Tautology or a Contradiction.
3. Show that $\sim(p \vee q) \rightarrow (\sim p \wedge \sim q)$ is a Tautology or a Contradiction.
4. Show that $[(\sim p \vee q)] \leftrightarrow (p \vee \sim q)$ is a Tautology or a Contradiction.
5. Show that $(p \wedge \sim q) \rightarrow \sim (p \wedge q)$ is a Tautology or a Contradiction.
6. Show that $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a Tautology or a Contradiction.
7. Examine whether the propositions $\sim(p \leftrightarrow q) \vee q \ and \ (p \wedge \sim q) \vee (q \wedge \sim p)$ are logically equivalent or not.
8. Examine whether the propositions $(p \wedge \sim q) \vee q \ and \ p \vee q$ are logically equivalent or not.
9. Examine whether the propositions $p \leftrightarrow q \ and \ (p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent or not.
10. Examine whether the propositions $p \leftrightarrow q \ and \ (p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent or not.

Chapter 7: RATIOS AND PROPORTIONS

One Mark Questions:

1. The ratio of 2hrs to 30mins is

   a) 4:1 b) 2:3 c) 2:1 d) None of these

2. The fourth proportional to $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ is

   a) 6 b) 4 c) 2 d) $\frac{1}{6}$

3. The fourth proportional to 1.5,4.5,3.5 is

   a) 10 b) 11 c) 10.5 d) 11.5

4. The third proportional to 2.4 and 3.6 is

   a) 5.4 b) 5 c) 6 d) 6.4

5. The mean proportional to $\frac{1}{16}$ and $\frac{1}{25}$

a) 10 b) 12 c) 15 d) None of these

6. The value of x if 5:20 = 3:x

a) 10 b) 12 c) 15 d) None of these

7. The mean proportional of 49 and 169

a) 91 b) 20 c) 8281 d) None of these

8. The fourth proportional of 6,14,15

a) 10 b) 20 c) 30 d) None of these

9. The sub triplicate ratio of $729 : 4913$

a) 17:9 b) 9:17 c) 3:17 d) None of these

10. Find the triplicate ratio of 3 : 5.

a) 27:125 b) 9:25 c) $\sqrt{3} : \sqrt{5}$ d) $\sqrt[3]{3} : \sqrt[3]{5}$

11. The duplicate ratio of 5 : 4

a) 25:16 b) $\sqrt{5} : \sqrt{4}$ c) 125:64 d) None of these

12. The compound ratio of 3 : 4 and 4 : 7

a) 3:7 b) 7:3 c) 21:16 d) None of these

13. The value of 'x' if $32 : x = 25 : 50$

a) 64 b) 128 c) 16 d) None of these

14. Find the sub duplicate of $225 : 9$.

15. Find the compound ratios of 1 : 2 and duplicate ratio of 2 : 3 and sub triplicate ratio of 125: 64.

16. Find the inverse ratio of 2 : 3.

17. A house consumes 20 kgs of rice and 5 kgs of wheat. Compare the consumption of rice and wheat in the form of a ratio.

18. The third proportional of 4 and 6 is

19. If a:b = 2:3 , b:c =5:7 , c:d = 3:1 then find a:d.

20. If a:b =3:4 and b:c = 5:12 then find a:b:c

Two Marks Questions:

1. Divide 6000 in the ratio 3 : 4 : 5.
2. 500Workers can finish a work in 8 days. How many workers will finish the same work in 5 days.
3. What must be added to each term in the ratio 2:3 so that it becomes 5:6.
4. What must be subtracted from each term in the ratio 7 : 4 so that it becomes 5 : 2 .
5. Two numbers are in the ratio 3 : 5. If 5 is added to each term, they are in the ratio 2 : 3. Find the numbers.
6. A ratio is in the lowest form given to be 3:8. If the difference between the quantities is 25. Find the quantities.
7. Find the ratio between two numbers such that their sum is 40 and their difference is 8.
8. If x:y = 2:3. Then find $\frac{2x^2+5y^2}{x^2+y^2}$.
9. If 10 persons can do a job in 30 days, then in how many days can 15 persons do the same job?
10. Find the numbers which when added to the terms of numerator and denominator of 25 : 37 will make it 5 : 6.
11. An article is sold at 40% gain on the cost price. Find the ratio of the selling price and cost price.

Three Marks Questions:

1. The monthly incomes of A and B are in the ratio 2:3 and their monthly expenditures are in the ratio 3:5. If each save Rs 100 per month , find the monthly incomes of A and B
2. X, Y and Z play cricket . The runs scored by X and Y are in the ratio 3:2. The ratio of the runs scored by Y and Z are in the ratio 3:2. Together they all score 342 runs. How many runs did each score?
3. Three numbers are in the ratio 2:3:4. If the sum of their squares is 1856, find the numbers

4. The monthly incomes of A and B are in the ratio 9:7 and those of B and C are in the ratio 3:2. If 10% of A's income and 15% of the C's income differ by Rs 18, Find the incomes of A,B and C
5. If a:b = 2:3 and x:y = 4:5 then find (5ax+3by): (10ax+4by)

Five Marks Questions:

1. Walking 4 kmph a student reaches his college 5 minutes late and if he walks at 5kmph then he reaches the college 2.5 minutes early. What is the distance from his house to the college?
2. Two taps can separately fill the tank in 12 min and 15 min respectively. The tank when full can be emptied by a drain pipe in 20 min. When the tank was empty all the three pipes were opened simultaneously, in what time will the tank be filled up?
3. 5Workers, each working 9 hours a day can finish a work in 30 days. How many workers are required to finish eight times the work in 25 days each working 8 hours a day?
4. A jar contains two liquids A and B in the ratio 7:5. When 9 liters of the mixture is drawn and the jar is filled with the same quantity of B, the ratio of A and B becomes 7:9. Find the quantity of A in the jar initially
5. 3 carpenters can earn Rs 360 in 6 days working 9 hours a day. How much will 8 carpenters can earn in 12 days working 6 hours a day?
6. Rs 5625 is divided among A, B and C so that A receives one half as much as B and C together receive and B receives one fourth of what A and C together receive . Find the shares of A, B and C
7. A railway train is 100 meters long and is running at the speed of 30kmph.In what time will it pass through
   i) a man standing on the platform and
   ii) a bridge which is 200 meters long?

Chapter 8: BILL DISCOUNTING

Two Marks Questions:

1. Find Banker's discount and True discount for a bill of Rs 415 due 9 months at 9% p.a.
2. Find the present value of Rs 900 due 4 months hence at the rate if 15% P.a. also find T.D
3. The Banker's gain on a bill is $1/9^{th}$ of the bankers discount, rate of interest is 10% find the unexpired period the bill
4. Find the present worth and the discounted cash value of the bill of Rs 2320, due 2 Years hence at 8% p.a.
5. Banker's discount and Banker's gain on a sum of money due 3 months hence are Rs 154.50 and Rs 150 respectively. Find the sum of money and rate of interest.
6. A Banker discounts a bill for a certain amount having 73 days to run before it matures at 15% p.a.The discounted value of the bill is Rs 970 What is the face value of the bill also find the Banker's discount
7. Banker's discount and Banker's gain on a bill due after sometime are Rs 1250 and Rs 50 Respectively. Find the face value of the bill.
8. A banker pays Rs 2380 on a bill of Rs 2500, 73 days before the legally due date. Find the rate of discount charged.
9. The difference between BD and TD on a certain sum of money due 6 months is Rs 27 Find the amount of the bill if the Rate of interest is 6% p.a.
10. True discount on a bill was Rs100 and Banker's gain is Rs 10 Find the Face value of the bill.

Three Marks Questions:

1. The difference between the Banker's discount and true discount on a bill due after 6 months at 4% interest p.a. is Rs 20. Find the true discount, banker's discount and the face value of the bill.

2. The bankers gain on a certain bill due 6 months hence is 10 the rate of interest being 10% p.a. Find the face value of the bill and the true present value.
3. A bill for Rs 14,600 drawn at 3 months after date was discounted on 11-11-99 for Rs 14320 if the discount rate is 20% p.a, on what date was the bill drawn?
4. A Bill for Rs 3500 due for 3 months was drawn on 27th march 2012 and was discounted on 18th April 2012, at a rate of 7% p.a. Find the bankers discount and discounted value of the bill.
5. A bill of Rs 12,900 was drawn on 3rd of February 2004 at 6 months and discounted on 13th march 2004 at 8% p.a. For what sum was the bill discount? And how much did the banker gain in this transaction?
6. The bankers gain on a bill is 1/15th of the banker's discount, rate of interest being 10% p.a. Find the unexpired period of the bill also find the true discount.
7. The present value of a bill due sometime hence is Rs 1100 and the true discount is Rs 110.Find the bankers discount.
8. A bill of Rs 1460 was drawn on 1stAugust for 6 months after date was discounted at 5% for Rs 1451.O/n what date was the bill discounted?
9. A bill of Rs 5000 drawn on 10/4/1998 at 12% p.a. for what sum was the bill discounted? And how much was the bankers gain?
10. A bill for Rs 12900 was drawn on 3rd February 2017 for 6 months and was discounted on 13th march 2017 at 8% p.a. Find the bankers present value and bankers gain?

Chapter 9: STOCK AND SHARES

<u>Two Marks Questions:</u>

1. What is the market value of 12% stock when an investment of Rs.6900 produces an income of Rs. 720?
2. How much of 8% stock can be purchased for Rs.4800? find the income obtained.

3. What income can be derived from an investment of Rs.10725 in 6.5% stock at 143? What is the amount of stock purchased?
4. What amount is realized by selling Rs.30000 stock at 20 discount?

Three Marks Questions:

1. A man invests equal sums of money in 4%, 5% and 6% stock, each stock being at par. If the total income of the man is Rs. 3600 Find his total Investment
2. Ramesh has invested Rs 4300 partly in 4.5% stock at Rs 72 and partly in 5% stock at Rs95. If the total income from both is Rs 250, find the investment in both the types of stock.
3. Sanjana invests Rs 3240 in a stock at 108 and sells when the price falls to 104. How much stock at 130 can Sanjana now buy?
4. Prathik sells out Rs 6000 of 7.5% stock at 108 and re-invests and proceeds in 9% stock. If Prathik's income increases by 270. At what price did Prathik buy 9% stock?
5. A person invests Rs 15000 partly in 3% stock at 75 and partly in 6% stock at 125. If the income from both is 675. Find his investment in the 2 types of stocks.
6. Rohan invested Rs 55000 partly in 8% stock at 80 and partly in 12% stock at 150 in such a way that to get a written of 9% for his money. How much did Rohan invest in each?
7. Sankeeth holds 2100 Rs of 3% stock. He sells at Rs 121 and invests then proceeds in 5% stock. Thereby his income increases by Rs 14. Find the market price of 5% stock.
8. Mr.Sandeep invests Rs 15000 cash partly in 3% stock at 75 and partly in 6% Debentures at 125 in such a way that to get a return on 4.5% for money. How much money does he need to invest? How much does he invest in each?
9. Rakshita invests 15000 in a company paying 7% per annum, when a share of value Rs 100 is selling for 150. What is her annual income and what percentage does she get on her money?
10. Which is a better investment 7% stock at 115 or 5% stock at 88.
11. What is the market value of 9.5% stock when an investment of Rs.12400 produces an income of Rs.1472.5?

12. What is the quoted value of 12% stock if it earns an interest of 8%after deducting the income tax of 8%?
13. A person ivested 42000 partly in 5% stock at 125 and the remaining in 7.5% stock at 75. If the income derived from two stocks is the same then find the investments in each type. Also find the total income.
14. Ayush buys Rs.10000 stock at 96 and sells when its price rises to Rs.102. find his gain.
15. Raksha invets in shares which are selling at Rs.2020 per share. How much money is required to purchase 10 shares if the brokerage is 0.5%?
16. Parashara sold 25 Biocon shares when the market price was Rs.400 per share. He then bought 10 Tata motors shares which were selling at Rs.800 per share. Brokerage for each transaction was 0.25%. The balance amount he gave to his daughter for travelling. How much did the daughter receive?

Chapter 10: LEARNING CURVE

Five Marks Questions:

1. An engineering company has 80% learning effect and spends 1000 hours to produce 1 lot of the product. Estimate the labour cost of producing 8 lots of the product, if the labour cost is Rs. 4 per hour.
2. A motor company limited has observed that a 90% learning effect applies to all labour related costs. Whenever a new product is taken up for production, the anticipated production is 320 units for the coming year. The production is done in lots of 10 units each. Each lot requires 1000 hours at $Rs.15$/hour. Calculate the total labour hours and labour cost to manufacture 320 units.
3. ABC Company required 1000 hours to produce first 30 engines. If the learning effect is90%. Find the total labour cost at $Rs.20$/hour to produce total of 120 engines.
4. An engineering company has 80% learning effect and spends 500 hours for the prototype. Estimate the labour cost of producing 7 engines of new order, if the labour cost is $Rs.40$ per hour.

5. An aircraft manufacturer supplies aircraft engines to different airlines. They have just completed an initial order for 30 engines involving a total of 6000 direct labour hours a5 Rs. 20 per hour. They have been asked to bid for a prospective contract for a supply of 90 engines. It is expected that there will be 80% learning effect. Estimate that labour cost for the New Year.
6. XYZ Company supplies water tankers to the government. The first water tanker takes 20,000 labour hours. The government auditors suggest that there should be 90% learning effect rate. The management expects an order of 8 water tankers in the next year, what will be the labour cost, the company will incur at the rate of Rs. 20 per hour.
7. A first sample batch of 50 units of product A took 80 hours to make. The company now wishes to estimate the average time per unit will be, if the total output of product A is 200 units and 80% learning rate applies.
8. A company requires 100 hours to produce the first 10 units at Rs. 15 per hour. The learning curve effect is 80%. Find the total labour cost to produce a total of 160 units.

Chapter 11: LINEAR PROGRAMMING PROBLEMS

<u>Five Marks Questions:</u>

Solve the following LPP graphically:

1. Maximize $z = 60x + 15y$
   Subject to the constraints $x + y \leq 50$, $3x + y \leq 90$ and $x \geq 0,\ y \geq 0$
2. Maximize $z = 5x + 3y$
   Subject to the constraints $3x + 5y \leq 15$, $5x + 2y \leq 10$ and $x \geq 0,\ y \geq 0$
3. Maximize $z = 0.08x + 0.10y$
   Subject to the constraints $3x + 5y \leq 15$, $5x + 2y \leq 10$ and $x \geq 0,\ y \geq 0$
4. Maximize $z = 6x + 8y$
   Subject to the constraints $4x + 2y \leq 20$, $2x + 5y \leq 24$ and $x \geq 0,\ y \geq 0$
5. Maximize $z = 2.5x + y$

Subject to the constraints $x + 3y \leq 12$, $3x + y \leq 12$ and $x \geq 0,\ y \geq 0$

6. Minimize $z = 50x + 75y$

   Subject to the constraints $2x + 3y \geq 6$, $3x + 4y \geq 8$ and $x \geq 0,\ y \geq 0$

7. Maximize $z = 2x + 3y$

   Subject to the constraints $x + y \leq 400$, $2x + y \leq 600$ and $x \geq 0, y \geq 0$

8. Maximize $z = 100x + 150y$

   Subject to the constraints $x + y \leq 50$, $2x + 2y \leq 85$ and $x \geq 0,\ y \geq 0$

9. Minimize $z = 3x + 8y$

   Subject to the constraints $x + y \geq 200$, $x \leq 8$, $y \geq 60$ and $x \geq 0,\ y \geq 0$

10. Maximize $z = 10500x + 9000y$

    Subject to the constraints $x + y \leq 50$, $2x + y \leq 80$ and $x \geq 0,\ y \geq 0$

11. Maximize $z = 400x + 200y$

    Subject to the constraints $5x + 2y \geq 30$, $2x + y \leq 15$, $x \leq y$ and $x \geq 0,\ y \geq 0$

Chapter 12: SALES TAX AND VALUE ADDED TAX

<u>Two Marks Questions:</u>

1. Abhishek purchase a bicycle costing ₹12,000. If the rate of sales tax is 9%. Calculate the total amount payable by him.
2. Ramu paid 60 as sales tax on a Titan Raga watch worth 1200. Find the rate of sales tax.
3. Refrigerator is marked for sale for 17,000 which include sales tax at 10% Calculate the sales tax in II. 3 Marks
4. Sharath paid 40 sales tax on a pair of shoes worth 500. Find the rate of sales tax?
5. The price of a washing machine inclusive of sales tax is 13,530. If the sales tax is 10%. Find the basic price
6. Gopal purchased a scooter costing 32,450. If the rate of sales tax is 9%. Calculate the total amount payable by him

7. A colour T.V is marked for sale for 17,600 which include sales tax at 10%. Calculate the sales tax in?
8. A shopkeeper purchase an article for 9,000 and sell it to a customer for 10,500. If the VAT rate is 4%. Find the VAT paid by the shopkeeper?
9. A shopkeeper purchase an article for 7,000 and sell it to a customer for 8,200. If the VAT rate is 6%. Find the VAT paid by the shopkeeper
10. Mr. Arya purchase an article for 3,100 and sell it to Mr. Aravind for 4,250. Mr. Aravind in turn sells it to Mr. Anil for 5,000. If the VAT levied 10%. Find the VAT levied on Arya and Aravind?

Three Marks Questions:

1. Bharath bought a shirt for Rs. 336 including 12% sales tax and a neck tie for Rs. 110 including 10% sales tax. Find the printed price of shirt and neck tie together.
2. A furniture dealers sold furniture for Rs. 21,000 and added 5% sales tax to the quoted price. The customer agrees to buy it for Rs. 21,000 including sales tax. Find the discount he received.
3. A shopkeeper sells an item at the price of Rs. 810 including ST of 8%. What should a customer pay for the same item, if the ST is reduced to 6%.
4. Sanju goes to a shop to buy a Bicycle quoted at Rs. 2,000. The rate of sales tax is 12% on it. He asks the shopkeeper for a rebate on the price of the Bicycle to such an extent that he has to pay Rs. 2,016 including sales tax. Find the rebate percentage on the price of the bicycle.
5. A shopkeeper announces a discount of 10% on a washing machine set. The marked priced of the washing machine is Rs. 22,000. How much will a customer have to pay for buying the washing machine set, if the rate of sales tax is 8%?
6. If the rate of sales tax is 5%, Sushma has to pay Rs. 7,140 for the steel cupboard. What amount she has to pay, if the sales tax is increased by 2%.

7. A shopkeeper purchases an audio system for Rs. 3,000 and sells it off at a gain of 15%. He also charges a sales tax of 10% on the selling price. Calculate the amount that the buyer will pay to the shopkeeper.
8. The price of T.V set inclusive of sales tax of 9% is Rs.13,407. Find its marked price. If the ST is increased to 13%, how much more does the customer Sneha pay for the T.V.?
9. When the rate of sales tax is decreased from 9% to 7%. For a Radio, Rahul has to pay Rs.632 less for it. What is the listed price of the radio?
10. Chandana purchases an article for Rs. 5,400 which includes 10% rebate on the marked price and 20% sales tax on the remaining price. Find the marked price of the article.
11. Mr. Ramesh buys a tape recorder for Rs.10,260 including sales tax. If the listed price of the tape recorder is Rs. 9,500. Find the rate of sales tax charged.
12. Rohith goes to purchase a motorcycle which is priced at Rs. 35,640 including 10% as sales tax. How was the actual rate of sales tax at the time of purchase is 7%. Find the extra profit made by the shopkeeper, if he still charges the original list price.
13. A shopkeeper bought a T.V at a discount of 30% of the listed price of Rs. 24,000. The shopkeeper offer a discount of 10% on the listed price to the customer. If the VAT is 10%. Find
    i) The amount paid by the customer.
    ii) The VAT to be paid by the shopkeeper.
14. A shopkeeper buys a mobile set at a discount of 20% from the wholesaler, the printed price of the mobile set being Rs. 1,600 and the rate of sales tax is 6%. The shopkeeper sells it to the buyer at the printed price and charges tax at the same time. Find
    i) The price at which the mobile set can be brought from the wholesaler.
    ii) The VAT to be paid by the shopkeeper.
15. ‘A’ is the manufacturer of electric iron. The cost price of each electric iron is Rs. 1,600. He sells to ‘B’ and ‘B’ sells it to ‘C’ and ‘C’ sells it to ‘D’ the retailer. The tax rate is 12.5% and the profit is Rs. 150 at each stage of the selling chain. Find
    i) The total amount of VAT.
    ii) The amount that the purchaser will have to pay.

Chapter 13: HEIGHTS AND DISTANCES

Four Marks Questions:

1. The angles of elevation of the top of a tower from the base and the top of a building are 60° and 30°. The building is 20 metres high. Find the height of the tower.
2. A person standing on the bank of a river observes that the angle subtended by a tree on the opposite bank is 60°. When he returns 40 metres from the bank, he finds the angle to be 30°. Find the height of the tree and the breadth of the river.
3. Two towers of height 14m and 25m stand on level ground. The angles of elevation of their tops from a point on the line joining their feet are 45° and 60° respectively. Find the distance between the towers.
4. From the top of a house 32m high, the angle of elevation of the top of a tower is 45° and the angle of depression of the foot of the tower is 30°.Find the height of the tower.
5. From a point A due north of the tower, the elevation of the top of a tower is 60°. From a point B due south, the elevation is 45°.If AB= 100metres. Show that the height of the tower is $50\sqrt{3}(\sqrt{3}-1)$ metres.
6. A flagstaff stands upon the top of a building. At a distance of 20metres, the angles of the top of the flag staff and building are 60° and 45° respectively. Find the height of the flagstaff.
7. The angles of elevation of the top of a tower from two points distant a and b (a<b) from its foot and the same the straight line from it are 30° and 60°. Show that the height of the tower is $\sqrt{ab}$.
8. From a point on the line joining the feet of two poles of equal heights, the angles of elevation of the tops of the poles are observed to be 30° and 60°. If the distance between the poles is a, find
   (i) The height of the poles
   (ii) The position of the point of observation

9. The angles of depressions of the two boats as observed from the mast head of a ship 50m high are 45° and 30°.What is the distance between the boats if they lie on the same side of the mast head in line with it.
10. A person is at the top of a tower 75feet high. From there, he observes a vertical pole and finds the angles of depression of the top and bottom of the pole which are $30^\circ$and 60°respectively. Find the height of the pole .
11. An aeroplane when flying at a height of 2000m passes vertically above another plane at an instant when their angles of elevation from the same point of observation are and respectively. Find the distance between the aeroplanes.
12. The angles of elevation of the summit of a hill from the top and the bottom of a tower are 30°and60°respectively. If the height of the tower is h, show that the height of the hill is $\frac{3h}{2}$.
13. The shadow of a tower standing on a level plane is found to be 50 metres longer when the Sun's altitude is 30°than when it is60°. Find the height of the tower.
14. A person at the top of a hill observes that the angles of depression of two consecutive kilometres stones on a road leading to the foot of the hill and in the same vertical plane containing the position of the observer are 30°and60°. Find the height of the hill.

Chapter 14: COMPOUND ANGLES, MULTIPLE AND SUBMULTIPLE ANGLES AND TRANSFORMATIOM FORMULAE

<u>One Mark Questions:</u>

Answer the following multiple choice questions

1. If $\sin A = \frac{3}{5}$, $\cos B = \frac{4}{5}$ and A,B are acute angles, then the value of $\sin(A+B)$ is

   a) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ b) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ c) $\frac{\sqrt{3}+1}{2}$ d) $\frac{\sqrt{3}-1}{\sqrt{2}}$

2. The value of $\sin 15^\circ$ is

a) $2-\sqrt{3}$ b) $2+\sqrt{3}$

c) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ d) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

3. The value of $\cos 75^{\circ}$ is

a) $\frac{1+\sqrt{2}}{2\sqrt{2}}$ b) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ c) $\frac{1}{2\sqrt{2}}$ d) $\frac{\sqrt{3}}{2\sqrt{2}}$

4. If $\tan A=\frac{1}{2}$, $\tan B=\frac{1}{3}$,then $\tan(A+B)$ is

a) 0 b) $\frac{1}{7}$ c) ∞ d) 1

5. If $\sin A=\frac{1}{2}$ then $\sin 2A$ is

a) $\frac{\sqrt{3}}{2}$ b) 0 c) 1 d) 2

6. If $\cos A=\frac{\sqrt{3}}{2}$ then $\cos 2A$ is

a) $2\sqrt{3}$ b) $\frac{2}{\sqrt{3}}$ c) $\frac{1}{2}$ d) 1

7. The value of $3\sin 10^{\circ}-4\sin^{3}10^{\circ}$ is

a) 10° b) 30° c) $\frac{\sqrt{3}}{2}$ d) $\frac{1}{2}$

8. The value of $4\cos^{3}10^{\circ}-3\cos 10^{\circ}$ is

a) 10° b) $\frac{\sqrt{3}}{2}$ c) 30° d) $\frac{1}{2}$

9. $\sin 4A.\cos 2A$ expressed as sum of two trigonometric functions is

a) $\frac{1}{2}(\sin 6A+\sin 2A)$ b) $\frac{1}{2}(\sin 6A+\cos 2A)$

c) $\frac{1}{2}(\cos 6A+\sin 2A)$ d) $\frac{1}{2}(\cos 6A+\cos 2A)$

10. $\cos 65^{\circ}.\cos 15^{\circ}$ expressed as sum of two trigonometric functions is

a) $\frac{1}{2}(\cos 70^{\circ}+\cos 50^{\circ})$ b) $\frac{1}{2}(\cos 80^{\circ}+\cos 50^{\circ})$

c) $\frac{1}{2}(\sin 70^{\circ}+\sin 50^{\circ})$ d) $\frac{1}{2}(\sin 80^{\circ}+\sin 50^{\circ})$

11. $2\cos 70°.\cos 10°$ expressed as sum of two trigonometric functions is

a) $\frac{1}{2}\left(\cos 80° + \frac{1}{2}\right)$

b) $\frac{1}{2}\left(\cos 80° + \frac{\sqrt{3}}{2}\right)$

c) $\cos 80° + \frac{1}{2}$

d) $\cos 80° + \frac{\sqrt{3}}{2}$

12. Match the following

i.	The value of $\cos 3A$ if $\cos A = \frac{4}{5}$ is	a. $\frac{1+\sqrt{3}}{2\sqrt{2}}$
ii.	If $\tan A = \frac{1}{\sqrt{3}}$, $\tan 2A$ is	b. $\frac{120}{169}$
iii.	If $\sin\theta = \frac{12}{13}$ and θ is acute, then $\sin 2\theta$ is	c. $\frac{1-\sqrt{3}}{2\sqrt{2}}$
iv.	The value of $\cos 105°$ is	d. $\sqrt{3}$
		e. $\frac{-44}{125}$

13. Express $\sin 5\theta + \sin\theta$ as product of two functions.

14. Express $\cos 4A.\sin 2A$ as sum or difference of two functions.

15. Express $\sin 80° - \sin 40°$ as product of two functions.

16. Write $\sin 12x + \sin 4x$ as product of two trigonometric functions.

<u>Five Marks Questions:</u>

1. Prove that $\frac{\cos 7x + \cos 3x - \cos 5x - \cos x}{\sin 7x - \sin 3x - \sin 5x + \sin x} = \cot 2x$

2. Prove that $\frac{\sin 2A + \sin 5A - \sin A}{\cos 2A + \cos 5A + \cos A} = \tan 2A$

3. Prove that $\frac{\sin 5A+\sin 4A+\sin 2A+\sin A}{\cos 5A+\cos 4A+\cos 2A+\cos A}=\tan 3A$

4. Show that $\frac{\sin 6A+\sin 2A+2\sin 4A}{\sin 7A+\sin 3A+2\sin 5A}=\frac{\sin 4A}{\sin 5A}$

5. If $A+B+C=180°$, prove that $\sin 2A+\sin 2B+\sin 2C=4\sin A.\sin B.\sin C$

6. Prove that $\cos A+\cos(120^{\circ}+A)+\cos\ (120^{\circ}-A)=0$.

7. Prove that $\tan A\,.\tan 3A\,.\tan 4A=\tan 4A-\tan 3A-\tan A$.

8. Prove that $\frac{1-\cos 2A+\sin 2A}{1+\cos 2A+\sin 2A}=\ \tan A$.

9. If A+B+C = π, prove that $\sin 2A+\sin 2B-\sin 2C=4\cos A\cos B\cos C$

Chapter 15: CIRCLES

<u>One Mark Questions:</u>

1. Find the equation of the circle with centre C and radius r in each of the following

   a) $c(-3,2)$ and $r=5$ units

   b) $c(-1,-2)$ and diameter $r=25$ units

   c) $c(acos\theta, asin\theta)$ and $r=a$ units

   d) $c(0,0)$ and $r=4$ units

   e) $c\ (1,1)$ and $r=\sqrt{2}$ units

2. Find the centre of the circle, two of the diameters are

   a) $x+y=2\ ,\ x-y=0$

   b) $y=0\ and\ y=x-5$

3. Write the equation of the point circle with centre at

   (a) (4,-5) (b) (-3, 2) (c) (1,0)

4. Find the centre and the radius of the circle.

   a) $x^2+y^2-4x-y-5=0$

   b) $(x-2)(x-4)+(y-1)(y-3)=0$

5. If the radius of the circle $x^2+\ y+4x-2y\ -k\ =0$ is 4 units find k.

6. Find the other end of the diameter, if one end of the diameter of the circle

   a) $x^2+y^2\ =\ 25\ is\ (5,0)$

   b) $x^2+y^2+4x-6y-12=0\ is\ (-5,-1)$

c) $x^2 + y^2 - 6x + 2y = 31$ is (7,4)

7. If (a, b) and (-5, 1) are the two end points of diameter of the circle $x^2 + y^2 + 4x - 4y = 2$. Find the values of a and b.
8. If $x^2 + y^2 - 4x - 8y + k = 0$ represents a point circle find k.
9. Show that the circle $x^2 + y^2 - 8x + 2y + 16 = 0$ touches x- axis
10. Show that the circle $x^2 + y^2 - 4x + 4y + 4 = 0$ touches x-axis

<u>Two Marks Questions:</u>

1. Find the equation of the circle. Two of the diameters are $x + y = 6$ and $x + 2y = 4$ and its radius is 10 units.

2. Two of the diameters $x + y = 4$ and $x - y = 2$ and passing through the point (2,-1)

3. Two of the diameters are $2x - 3y = 5$ and $3x - 4y = 7$ are the diameters of a circle of area 154 sq.cm

4. Find the equation of the circle with centre at (-2, 1) and passing through the origin.

5. Find the equation of the circle with centre at (2,1) and passing through (0,-1)

6. Find the equation of the circle described on the line joining the points A and B as diameter where

a) A(-5, 1) and B(1,3) b) A(2,0) and B(0,2) c)A(3,4) and B(1,-2)

7. Find the equation of the circle whose centre is same as the centre of the circle $x^2 + y^2 + 6x + 2y + 1 = 0$, and passing through the point (-2,-1).

8. Find the equation of the circle whose centre is same as the centre of the circle $x^2 + y^2 - 6x + 4y + 9 = 0$ and passing through the point (-2,3).

9. Find the equation of the circle whose centre is (-2, 3) and passing through the centre of the circle $x^2 + y^2 - 6x + 4y + 9 = 0$.

10. Find the equation of the circle passing through the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ and centre at (4,-2).

11. Find the equation of the circle two of whose diameters are $x + y = 3$ and $2x + y = 2$ and passing through the centre of the circle $x^2 + y^2 - 4x + 2y - 1 = 0$.

12. Find the equation of the circle concentric with the centre of the circle $x^2 + y^2 - 2x + 2y - 1 = 0$ and having double its area.

13. Find the equation of the circle concentric with the centre of the circle

14. Find the length of the chord of the circle $x^2 + y^2 - 6x + 4y + 5 = 0$ intercepted by x – axis

15. Find the length of the chord of the circle $x^2 + y^2 - 6x + 15y - 16 = 0$ intercepted by the y-axis.

16. Find the length of the chord of the circle $x^2 + y^2 + 3x - 2 = 0$ intercepted by y -axis.

<u>Five Marks Questions:</u>

1. Find the equation of the circle passing through the points $(5, 3), (1, 5)$and $(3, -1)$.
2. Find the equation of the circle passing through the points $(5, 7), (6, 6)$and $(2, -2)$.
3. Find the equation of the circle passing through the points $(1, 1), (-2, 2)$and $(-6, 0)$.
4. Find the equation of the circle passing through the points $(0, 2), (3, 0)$and $(3, 2)$.
5. Find the equation of the circle passing through the points $(1, 1), (5, -5)$and $(6, -4)$.
6. Find the equation of the circle passing through the points $(0, 1), (2, 3)$and $(-2, 5)$
7. Find the equation of the circle passing through the points $(1, -4)$ and $(5, 2)$ and having its centre on the line $x - 2y + 9 = 0$.
8. Find the equation of the circle passing through the points $(5, 1)$ and $(3, 4)$ and has its centre on X-axis.
9. Find the equation of the circle passing through the points $(1, 2)$ and $(2, 1)$ and has its centre on Y-axis.
10. Find the equation of the circle passing through the points $(-1, 2)$ and $(3, -2)$ and has its centre on $x = 2y$.
11. Find the equation of the circle passing through the point $(2, 3)$ having its centre on X-axis and radius 5 units.

12. A circle has radius 3 units and its centre lies on the line $y = x - 1$. Find the equation of the circle, if it passes through $(7, 3)$.
13. Find the equation of the circle passing through $(1, 1)$ and $(2, 2)$ and having radius 1 unit.
14. Find the value of k, for which the line $x + ky - 5 = 0$ may touch the circle $x^2 + y^2 - 2x - 6y - 6 = 0$.

Six Marks Questions:

1. Show that the points $(2, 0), (-1, 3), (-2, 0)$and $(1, -1)$ are concyclic.
2. Show that the points $(0, 0), (1, 1), (5, -5)$and $(6, -4)$ are concyclic.
3. Show that the points $(2, -4), (3, -1), (3, -3)$and $(0, 0)$ are concyclic.
4. Check whether the points $(1, 0), (2, -7), (8, 1)$and $(9, -6)$ are concyclic or not.

Chapter 16: PARABOLA

One Mark questions:

1. The focus of the parabola $x^2 = 16y$ is
   a) (0, 4) b) (4,0) c) (16,0) d) (0,16)
2. The vertex of the parabola $x^2 = y$ is
   a) (0, 0) b) (1,0) c) (-1,0) d) (0,1)
3. The length of the latus rectum of the parabola $y^2 = 8x$ is
   a) 8 b) 4 c) 16 d) 2
4. The axis of the parabola $x^2 = 5y$ is
   a) y axis b) x axis c) origin d) y=0
5. The equation of the latus rectum of the parabola $y^2 = 4ax$ is
   a)x +a =0 b) x-a =0 c) y+a =0 d) y-a =0

Two Marks questions:

1. Find the equation of the parabola given that its vertex is (0,0) and focus is (4,0).
2. Find the equation of the parabola given that its vertex is (0,0) and the directrix is y=-3.

3. Find the equation of the parabola given that its focus is (0,1) and directrix is y= -1.
4. Find the equation of the parabola whose focus is (0,-3) and its directrix is y= 3.
5. Find the equation of the parabola whose focus is (0,6) and its vertex is (0,0).
6. Find the equation of the parabola given that its vertex is (0,0) and focus is (-3, 0).
7. Find the equation of the parabola given that its vertex is (0,0) and focus is (0,-5/2).
8. Find the equation of the parabola given that the focus is (5/3, 0) and vertex is (0,0).
9. Find the equation of the parabola given that the focus is (0,-3/5) and the directrix is 5y = 3.
10. Find the equation of the directrix, length of the latus rectum of the parabola y^2= -6x.
11. Find the vertex and focus of the parabola y^2 = 12x.
12. Find the equation of the latus rectum, length of the latus rectum of the parabola x^2= -24y.
13. Find the equation of the axis and the length of the latus rectum of the parabola x^2 = 15y
14. Find the focus, equation of latus rectum of the parabola y^2 = 4x.
15. Find the equation of the axis, equation of the directrix of parabola x^2 + 8y = 0.
16. Find the axis and the length of latus rectum of the parabola x^2= 16y.
17. Find the end points of latus rectum of the parabola y^2 = 18x.
18. Find the value of 'k' if the length of the latus rectum x^2 = -2ky is 8.
19. Find the equation of the parabola whose vertex is (0,0), focus is (-4,0) and equation of the directrix is x = 4.
20. Find the equation of the parabola whose focus is (0,-4) and the vertex is (0,0).

<u>Three Marks Questions:</u>

1. Find the equation of the parabola given that its vertex is (0,0) and passing through the point (3,-4) and symmetric about the y axis
2. Find the equation of the parabola given that its vertex is (0,0) and the axis along the x axis and passing through the point P(2,3).
3. Find the equation of the parabola given that its vertex at the origin , axis is the y axis and passing through the point (1,3)

Chapter 17: LIMITS AND CONTINUITY

One Mark Questions:

1. Evaluate: $\lim_{x\to 3}\left(\frac{x^2-4x}{x-2}\right)$.
2. Evaluate: $\lim_{x\to 0}\frac{\sin ax}{bx}, a, b \neq 0$.
3. Evaluate: $\lim_{x\to 0}\frac{\sin 4x}{\sin 2x}$.
4. Evaluate: $\lim_{x\to 0}\frac{\sin x^0}{x}$.
5. Evaluate: $\lim_{x\to 0}\frac{\sin x^2}{x}$.
6. Evaluate: $\lim_{x\to 4}\left(\frac{4x+3}{x-2}\right)$.
7. Evaluate: $\lim_{x\to 1}\left(\frac{ax^2+bx+c}{cx^2+bx+a}\right)$.
8. Evaluate: $\lim_{x\to 1}(x^2+6x+4)$.
9. Evaluate: $\lim_{x\to 2}\left(\sqrt{36-x^2}\right)$.
10. Evaluate: $\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^n$.
11. Evaluate: $\lim_{x\to 0}(1+3x)^{\frac{1}{x}}$.
12. Evaluate: $\lim_{n\to 0}\left(\frac{n+3}{3}\right)^{\frac{2}{n}}$.
13. Evaluate: $\lim_{x\to 0}\left(\frac{3^x-2^x}{x}\right)$.
14. Evaluate: $\lim_{x\to 0}\left(\frac{2^x-1}{3x}\right)$.
15. Evaluate: $\lim_{x\to 0}\left(\frac{e^{-x}-1}{x}\right)$.

Two Marks Questions:

1. Evaluate: $\lim_{x\to 0}\frac{x\cos x+\sin x}{x^2+\tan x}$.
2. Evaluate: $\lim_{x\to 0}(\operatorname{cosec} x-\cot x)$.
3. Evaluate: $\lim_{x\to 0}\frac{x\tan 4x}{1-\cos 4x}$.
4. Evaluate: $\lim_{x\to 0}\frac{3^{2x}-1}{2^{3x}-1}$.
5. Evaluate: $\lim_{x\to\infty}\frac{3x^2-4x+5}{2x^2+5x-1}$.
6. Evaluate: Show that the function $f(x)=\begin{cases}(1+3x)^{\frac{1}{x}} & x\neq 0\\ e^3 & x=0\end{cases}$ is continuous at x = 0.
7. Evaluate: Find k for which $f(x)=\begin{cases}k+x, & x=1\\ 4x+3, & x\neq 1\end{cases}$ iscontinuous at x=1.
8. Evaluate: If the function $f(x)=\begin{cases}(1+2x)^{\frac{1}{x}}, & x\neq 0\\ k, & x=0\end{cases}$ is continuous at x=0, find k.

9. Evaluate:Find k for which $f(x) = \begin{cases} \frac{e^{2x}-1}{x} & x \neq 0 \\ k & x = 0 \end{cases}$ is continuous at x=0.

Three Marks Questions:

1. Evaluate: $\lim_{x\to 3}\left[\frac{x^2-4x+3}{x^2-2x-3}\right]$.
2. Evaluate: $\lim_{x\to 0}\left[\frac{\sqrt{1+x}-\sqrt{1-x}}{x}\right]$.
3. Evaluate: $\lim_{\theta\to 0}\left[\frac{1-\cos 4\theta}{1-\cos 6\theta}\right]$.
4. Evaluate: $\lim_{x\to 0}\left[\frac{\sin 2x+\sin 6x}{\sin 5x-\sin 3x}\right]$.
5. Evaluate: $\lim_{x\to 0}\frac{9^x+9^{-x}-2}{x^2}$.
6. Evaluate: $\lim_{x\to\infty}\left(\frac{x-3}{x+3}\right)^{x+2}$.
7. Evaluate: $\lim_{x\to\infty}\sqrt{x}\left(\sqrt{x+3}-\sqrt{x}\right)$.
8. Evaluate: $\lim_{x\to\infty}x\left(\sqrt{x^2+1}-x\right)$.
9. Evaluate: Prove that $f(x) = \begin{cases} x^2+1 & when \quad x<2 \\ 5 & when \quad x=2 \\ 4x-3 & when \quad x>2 \end{cases}$ is continuous at x = 2.
10. Evaluate: Discuss the continuity of $f(x) = \begin{cases} 3x^2+1 & if \quad x<1 \\ 4 & if \quad x=1 \\ 2x+2 & if \quad x>1 \end{cases}$ at x = 1.

Six Marks Question:

1. If n is a rational number and A is a non-zero real number then prove that

$\lim_{x\to a}\left[\frac{x^n-a^n}{x-a}\right] = na^{n-1}$.

Chapter 18: DIFFERENTIAL CALCULUS

Multiple Choice Questions:

1. The derivative of $\log e^e$ w.r.t. x is
   a) 1 b) 0 c) e^e d) $1/e^e$
2. If y= $\sin(tanx)$ $\frac{dy}{dx}$ is
   a)Cos(tanx) b) cosx. Sec^2x c) Cos(tanx) Sec^2x d) - Cos(tanx) Sec^2x
3. If x-y=0 then $\frac{dy}{dx}$ is
   a) 1 b) 0 c) -1 d) x
4. If $f(x)=x^2+2x+3$ the $f^1(1)$ is

a) 6 b) 5 c) 8 d) 4

5. The derivative of $\sqrt{x}$ is

a) $\frac{1}{\sqrt{x}}$ b) $2\sqrt{x}$ $c)$ $\frac{1}{2\sqrt{x}}$ d) $\frac{1}{2}\sqrt{x}$

6. The derivative of $\sin^2 x$ is

a) $\text{Cos}^2 x$ b) 2sinx c) 2 cosx d) sin2x

7. If y= log(sinx) then $\frac{dy}{dx}$ is

a) cosecx b) cosx c) cotx d) tanx

8. If y= $\sin^2 x + \text{Cos}^2 x$ then $\frac{dy}{dx}$ is

a) sin2x b) sinx +cosx c) 1 d) 0

9. If y=2^x then $\frac{dy}{dx}$ is

a) 2^x b) $x\, 2^{x-1}$ c) $2^x \log 2$ d) $2^x \log 2$

10. If y= $\frac{4x^2-3x}{x}$ $then$ $\frac{dy}{dx}$

a) 8x – 3 b) 4 c) $\frac{4x^2-3x}{x^2}$ d) 4x-3

<u>One Mark Questions:</u>

1. If y=$x^e + e^x + e^e$ Find $\frac{dy}{dx}$
2. Differentiate the following w.r.t. x
   $5e^x - logx - 3\sqrt{x}$
3. If y= $(\sqrt{x} + \frac{1}{\sqrt{x}})^2$ Find $\frac{dy}{dx}$
4. If y = $(3x^2 + 4x + 5)^6$ $Find$ $\frac{dy}{dx}$
5. If y= e^{x^2} Find $\frac{dy}{dx}$
6. If y= log $(x^2 - 2)$ Find $\frac{dy}{dx}$
7. Differentiate $tan^4 x$ $w.r.t.$ x
8. If y = sec(secx) Find $\frac{dy}{dx}$
9. If y =($a^2 - x^2)^{10}$ Find $\frac{dy}{dx}$
10. If y = log (log (logx) Find $\frac{dy}{dx}$
11. If y = cos x^3 Find $\frac{dy}{dx}$
12. If y = $\sqrt{cot\sqrt{x}}$ Find $\frac{dy}{dx}$
13. If y = log(sin $\sqrt{x}$)) Find $\frac{dy}{dx}$
14. If y = $sin^3\sqrt{x}$ Find $\frac{dy}{dx}$
15. If $x^2 + y^2 = a^2$ $Find$ $\frac{dy}{dx}$
16. If $\sqrt{x} + \sqrt{y} = 3$ $Find$ $\frac{dy}{dx}$
17. If $y^2 = 4ax$ $Find$ $\frac{dy}{dx}$
18. If $x^2 - y^2 = a^2$ $Find$ $\frac{dy}{dx}$
19. If $x^2 = 4ay$ $Find$ $\frac{dy}{dx}$
20. $3x^2 + 4y^2 = 10$ $Find$ $\frac{dy}{dx}$
21. If y = $\frac{x+1}{x}$ $Find$ $\frac{dy}{dx}$
22. If $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ $Find$ $\frac{dy}{dx}$
23. If y = $\frac{2}{x+1}$ Find $\frac{dy}{dx}$
24. If y = (x-a) (x-b) Find $\frac{dy}{dx}$
25. If y = $\frac{x-a}{x-b}$ Find $\frac{dy}{dx}$
26. If y = $sin^2 x$ Find $\frac{dy}{dx}$

27.If y = $2^{6log_2 x}$ Find $\frac{dy}{dx}$

28.If y = sec x · tan x Find $\frac{dy}{dx}$

29.If y = $x^9 \cdot 9^x$ Find $\frac{dy}{dx}$

30.If y = $2^x - x^2 - 3logx$ Find $\frac{dy}{dx}$

31. If y= $\frac{3x^2+2x+5}{\sqrt{x}}$ Find $\frac{dy}{dx}$

32.If $y = \frac{1}{x^{4/3}} - \frac{3}{x^{3/2}}$ Find $\frac{dy}{dx}$

33.If $y = x^2 + \frac{4}{x^2} - \frac{2}{3}tanx + 6e$ Find $\frac{dy}{dx}$

34.If $y = \sqrt{tan\sqrt{x}}$ Find $\frac{dy}{dx}$

35.If $y = log(secx + tanx)$ Find $\frac{dy}{dx}$

36.If $y = tan(logsinx)$ Find $\frac{dy}{dx}$

37.If $y = sinx.Sin2x$ Find $\frac{dy}{dx}$

38.If $y = e^{2x}.Sin3x$ Find $\frac{dy}{dx}$

39.If $y = log(log(tanx))$ Find $\frac{dy}{dx}$

40.If $y = 7^{Sin\sqrt{x}}$ Find $\frac{dy}{dx}$

41.If $y = [log(cosx)]^2$ Find $\frac{dy}{dx}$

<u>Two Marks Questions:</u>

1. Differentiate x^x w.r.t. x
2. y= $\sqrt{x + \sqrt{x + \sqrt{x + \cdots \infty}}}$ Prove that $\frac{dy}{dx} = \frac{1}{2y-1}$
3. y= $\sqrt{tanx + \sqrt{tanx + \sqrt{tanx + \cdots \infty}}}$ Prove that $\frac{dy}{dx} = \frac{sec^2x}{2y-1}$
4. y= $\sqrt{sinx + \sqrt{sinx + \sqrt{sinx + \cdots \infty}}}$ Prove that $\frac{dy}{dx} = \frac{cosx}{2y-1}$
5. Differentiate($log^x)^{cosx}$ w.r.t x
6. Differentiate x^{sinx} w.r.t x
7. If $x^y = e^{x-y}$ Prove that $\frac{dy}{dx} = \frac{logx}{(1+logx)^2}$
8. If x= a cosθ , y=a sinθ Find $\frac{dy}{dx}$
9. If x=at^2 , y=2at Find $\frac{dy}{dx}$
10. If x = aθ , y=$\frac{a}{\theta}$ Prove that $\frac{dy}{dx} + \frac{y}{x} = 0$
11. Differentiate sin^3x w.r.t cos^2x
12. If x=log t , y=$\frac{1}{t}$ Find $\frac{dy}{dx}$
13. If x = a secθ , y= b tanθ Find $\frac{dy}{dx}$
14. If x=4t , y=$\frac{4}{t}$ Find $\frac{dy}{dx}$
15. Differentiate tan$\sqrt{x}$ w.r.t $\sqrt{x}$
16. Differentiate logx w.r.t $\frac{1}{x}$
17. If $x^y = y^x$ Find $\frac{dy}{dx}$

18.If $\frac{x^2}{a^2}+\frac{y^2}{b^3}=1$. Find $\frac{dy}{dx}$

19.If $x^3+y^3=3axy$. Find $\frac{dy}{dx}$

20.If y=log (x + $\sqrt{1+x^2}$) Prove that $\frac{dy}{dx}=\frac{1}{\sqrt{1+x^2}}$

21.If y = $\sqrt{\frac{1-cos2x}{1+cos2x}}$ Prove that $\frac{dy}{dx}=sec^2x$

22.If f(x) =$x^2-3x+10$ Find $f^{|}(50)$

23.If f(x)=x^n and $f^{|}(1)=10$. Find n

24.If y= x + $\frac{1}{x}$ Show that $\frac{x^2dy}{dx}-xy+2=0$

25.If y= $x^{x^{\cdot^{\cdot^{\infty}}}}$ Prove that $\frac{dy}{dx}=\frac{y^2}{x(1-ylogx)}$

26.Differentiate x^x w.r.t. x

27.Differentiate $x^{\sqrt{x}}$ w.r.t. x

28.Differentiate $(sinx)^x$ w.r.t. x

29.Differentiate $(sinx)^{tanx}$ w.r.t. x

30.x= log(1+t) , y=$\frac{1}{1+t}$ Find $\frac{dy}{dx}$

Three Marks questions:

1. $x^y=y^x show\ that\ \frac{dy}{dx}=\frac{y(y-xlogx)}{x(x-ylog\ x)}$
2. If $x=e^{logcos4\theta}y=e^{logsin4\theta}$show that $\frac{dy}{dx}=\frac{-x}{y}$
3. If $x=acos^4\theta\quad y=asin^4\theta$ show that $\frac{dy}{dx}=-tan^2\theta$
4. If $x=e^t(cost+sint)\quad y=e^t(cost-sint)$ show that $\frac{dy}{dx}=-tant$
5. If $x=a\ \log(sec\theta)\quad y=a(tan\theta-1)$ show that $\frac{dy}{dx}=-2Cosec2\theta$
6. Differentiate $logx\ w.r.t.\frac{1}{x}$
7. Differentiate $logsinx\ w.r.t.\sqrt{cosx}$
8. Differentiate $tan\sqrt{x}\ w.r.t.\sqrt{x}$
9. If $x=a\left[cost+logtan\frac{t}{2}\right]\quad y=a\ sint\ \ show\ that\ \frac{dy}{dx}=tant$
10. Differentiate $tan^2x\ w.r.t.cos^2x.$

Five Marks Questions:

1. If $y=aCos(logx)+bsin(logx)\ Prove\ that\ x^2y_2+xy_1+y=0$

2. If $y = \log(x - \sqrt{x^2} + 1)$. Show that $(x^2 + 1)y_2 + xy_1 = 0$
3. If y=$x + \sqrt{x^2 - 1}$. Show that $(x^2 - 1)y_2 + xy_1 - y = 0$
4. If y=$(x + \sqrt{x^2 + 1})^m$.S how that $(x^2 + 1)y_2 + xy_1 - m^2y = 0$
5. If y=sin(logx) . Show that $(x^2y_2 + xy_1 + y = 0)$
6. If y=a cos m x + b sin m x. Show that $\frac{d^2y}{dx^2}$ + $m^2y = 0$
7. If y= $(a^2 + x^2)^6$. Show that $(x^2 + a^2)y_2 - 10xy_1 - 12y = 0$
8. If xy + 6y = 2x. Show that $\frac{d^2y}{dx^2}$ = $\frac{-24}{(x+6)^3}$
9. If y= $e^x logx$ Show that $xy_2 - (2x - 1)y_1 + (x - 1)y = 0$
10. If y=log (x +$\sqrt{x^2 + 1)}$. Show that $(x^2 + 1)y_2 + xy_1 = 0$

Chapter 19: APPLICATION OF DERIVATIVES

<u>Two Marks Questions:</u>

1. If $S = t^3 - 6t^2 + 9t + 8$. Find the initial velocity.
2. A particle moves a distance $S = 6t^2 - t^3 + 5$ then find the velocity and acceleration.
3. The distance S travelled by a particle in time t seconds is given by $S = t^3 - 6t^2 + 15t + 2$, find when the acceleration is zero.
4. If the displacement S at any time t is given by $S = \sqrt{1 - t}$. Show that the velocity is inversely proportional to the displacement.
5. If the displacement S at any time t is given by $S = 2t^3 - 5t^2 + 4t - 3$. Find the time, when acceleration is 14ft/sec^2.
6. If the displacement S at any time t is given by $S = 2t^3 - 5t^2 + 4$. Find the velocity and displacement at that time.
7. When brakes are applied to a moving car, the car travels a distance S feet in t seconds given by $S = 20t - 40t^2$. When and where the car stops?
8. A particle moves according to the laws $S = t^3 + at^2 + bt$. Find a and b, if initial velocity is 5units and when $t = 1$ sec, it is moving with a velocity which is 4 times its initial velocity.
9. A particle shot vertically upwards rises ‘s’ feet in ‘t’ sec, where $S = 40t - 16t^2$. Find the greatest height attained by the particle.

10. The displacement S of a particle at time t is given by $S = 4t^3 - 6t^2 + t - 7$. Find the velocity and acceleration, when $t = 2\ sec$.
11. If $S = 5t^2 + 4t - 8$. Find the initial velocity and acceleration.
12. A stone thrown vertically upward rises 's' feet in 't' sec, where $S = 80t - 16t^2$. What will be the velocity and acceleration after 2 secs.
13. A body is thrown vertically upwards, its distance s feet in t sec is given by $S = 5 + 12t - t^2$. Find the greatest height reached by the body.
14. If $S = at^3 + bt$, find 'a' and 'b' given that when $t = 3sec$, velocity is 0 and acceleration is 14 units.
15. When brakes are applied to a moving car, the car travels a distance S feet in t seconds given by $S = 8t - 6t^2$. When and where the car stops?
16. If the cost function of a firm is given by $C(x) = x^3 - 3x + 7$. Find the average cost and marginal cost, when the output is $x = 6$ units.
17. The total cost function is given by $q^3 - 3q^2 + 15q + 27$. Find the average cost and marginal cost.
18. If total cost function is given by $C = 9q - 3\mathrm{q}^2 + \frac{\mathrm{q}^3}{3}$. Find the level of output at which average cost is minimized.
19. The total cost of a commodity is given by $C = x^2 - 7x + 2$, where x is the number of units and price/unit is Rs. 5.00. find the profit function.
20. Find the average cost and marginal cost, If total cost function of an article is given by $C = 5\mathrm{x}^2 + 2\mathrm{x} + 3$.
21. The total cost of a commodity is given by $C = -x^2 + 5x + 7$ and price/unit is 12.00rs. Find the profit function.
22. For the demand function $2x - 5y = 7$. Find the total revenue, marginal revenue and average revenue.
23. The total cost C for output Q is given by $C = 300Q - 10\mathrm{Q}^2 + \frac{\mathrm{Q}^3}{3}$. Find the level of output at which marginal cost is minimized.

24. If $R = 250x + 45x^2 - x^3$. What will be the marginal revenue, if $x = 25units$ and what is average revenue, when $x = 10\ units$.
25. The sum of 2 natural numbers is 48. Find them, if their product is maximum.
26. The product of 2 numbers is 64. Find the numbers, when their sum is minimum.
27. The total revenue function is given by $R = 400x - 2x^2$ and the total cost function is given by $C = 2x^2 + 40x + 4000$. Find the marginal revenue and marginal cost function.
28. If the total cost $C(x) = x^2 + 2x + 1$ then find the marginal cost and average cost.

<u>Three Marks Questions:</u>

1. The sides of an equilateral triangle is increasing at the rate of $\sqrt{3}cm/sec$. Find the rate at which its area is increasing, when its side is 200cms.
2. The surface area of a spherical bubble is increasing at the rate of $0.6cm^2/sec$. Find the rate at which its volume is increasing, when its radius is 3cm.
3. A circular patch of oil spreads on water, the area growing at the rate of $16cm^2/min$. How fast are the radius and the circumference increasing, when the diameter is 12cms?
4. The radius of a circular plate increases at the rate of $\frac{8}{3\pi}\ cm/sec$. Find the rate of change in the area, when the diameter is 12cm. Also find the rate of increase of circumference of the circle after 3secs.
5. The volume of a sphere is increasing at the rate of $4\pi\ c.c/sec$. Find the rate at which area of its surface increases, when the radius is 10cm.
6. A spherical balloon is being inflated so that its volume is increasing at the rate of $30\ c.c/min$. How fast its surface area increasing, when its volume is $36\pi\ c.c$.
7. A square plate is expanding uniformly, the side is increasing at the rate of $5cm/sec$. What is the rate at which the area and its perimeter is increasing, when the side is 20cm long.
8. The sides of an equilateral triangle are increasing at the rate of $3\ cm/sec$. How fast is its area increasing, when the side is 10 cm.
9. The edge of a variable cube is increasing at the rate of $6\ cm/min$. How fast is the volume and its surface area increasing when the edge is 10 cm long.

10. A ladder 17 feet long leaning against a smooth vertical wall. If the lower end is moving at the rate of $12\ ft/min$. Find the rate at which the upper end is moving, when the lower end is $8ft$ from the wall.
11. The radius of a sphere is increasing at the rate of $0.5\ mt/sec$. Find the rate of increase of its Surface area and volume after 3 sec.
12. The surface area of a spherical bubble is increasing at the rate of $0.8\ cm^2/sec$. Find at what rate is its volume increasing when $r = 2.5cm$.
13. A spherical balloon is being inflated at the rate of $35cc/sec$. Find the rate at which the surface area of the balloon increases, when its diameter is 14 cm.
14. The radius of circular plate is increasing at the rate of $\frac{2}{3\pi} cm/sec$. Find the rate of change of its area, when the radius is 6 cm.
15. Water is being poured at the rate of $30mt^3/min$ into a cylindical vessel whose base is a circle of radius 3mts. Find the rate at which the level of water is rising.
16. Sand is being dropped at the rate of $10mt^3/sec$ into a conical pile. If the height of pile is twice the radius of the base. At what rate is the height of the pile increasing, when the sand in the pile is 8mt high.
17. A man 6 ft tall is moving directly away from a lamp post of height 10 ft above the ground. If he is moving at the rate of 3 ft/sec. Find the rate at which the length of his shadow is increasing and also tip of shadow is moving.
18. The volume of a spherical ball is increasing at the rate of $4\pi\ c.c/sec$. Find the rate of increase of the radius of the ball, when the volume is $288\pi\ c.c$.
19. A drop of ink spreads over a blotting paper so that the circumference of the blot is $4\pi\ cm$ and it changes at $3cm/sec$. Find the rate of increase of its radius and also find rate of increase of its area.
20. A circular plate of metal is heated so that its radius increases at the rate of $0.1\ mm/min$. At what rate is the plates area increasing, when the radius is $25\ cm$ $(1cm = 10m)$.
21. Find the minimum value of $x^2 + \frac{250}{x}$
22. Find maximum and minimum value of $x^3 - 9x^2 + 15x - 1$

23. Find maximum and minimum value of the function $f(x) = 3x^3 - 9x^2 - 27x + 30$

24. Find the maximum and minimum value of $f(x) = x^5 - 5x^4 + 5x^3 - 1$

25. Show that $x^3 - 6x^2 + 12x - 3$ has neither a maximum nor a mininmum at $x = 2$

26. Divide the number 40 into two parts such that their product is maximum.

27. The product of two natural number is 144, find the numbers if their sum is minimum.

28. Divide 64 into two parts such that the sum of the cubes of two parts in minimum.

29. Find the maximum and minimum of the function $f(x) = x^3 - 12x^2 + 36x - 4$

30. Find the maximum and minimum value of the function $f(x) = 2x^3 - 3x$

31. Find the maximum and minimum value of the function

$f(x) = x^3 - 6x^2 + 9x + 15 \quad (0 \le x \le 6)$.

32. Find the maximum and minimum value of the function $f(x) = x^4 - 62x^2 + 120x + 9$

33. Find the maximum and minimum value of the function $f(x) = 2x^3 - 3x^2 + 312x + 12$

34. Find the maximum and minimum value of the function $f(x) = 2x^3 - 3x^2 + 36x + 10$

35. Find the maximum and minimum value of the function $f(x) = 9x^3 + 12x + 2$

Chapter 20: INDEFINITE INTEGRALS

<u>Answer the following multiple choice questions:</u>

1. The value of $\int e^{3x}dx$ is _____

a) e^{3x} b) $3e^{3x} + c$ c) $\frac{e^{3x}}{3} + c$ d) $\frac{e^{3x}}{3}$

2. $\int \frac{x^4 + 3x^2 - 5x}{x^2}dx$ is equal to _____

a) $\frac{x^3}{3} + 3x - 5\log x + c$ b) $3x^2 + 3x - \log x + c$ c) $x^3 + 3x - 5\log x + c$ d) $\frac{x^2}{2} + 3 - \log x + c$

3. $\int \left(x^e + e^x - \log_e a\right)dx$ is equal to _____

a) $e\left(x^{e-1}\right) + e^x + c$ b) $\frac{x^{e+1}}{e+1} + e^x - 1 + c$

c) $\frac{x^{e+1}}{e+1}+e^x-x\log_e a+c$ d) x^e+e^x-1+c

4. $\int \frac{1}{7x+8}dx$ is equal to _____

a) $\log|7x+8|+c$ b) $\frac{\log|7x+8|}{7}+c$ c) $\frac{\log|7x+8|}{8}+c$ d) $\log\left|\frac{7x+8}{7}\right|+c$

5. The value of $\int e^{5x+7}dx$ is _____

a) $e^{5x}+c$ b) $e^{5x+7}+c$ c) $5e^{5x+7}+c$ d) $7e^{5x+7}+c$

6. $\int \tan x dx$ is _____

a) log(sin x)+c b) log(cot x)+c c) log(sec x)+c d) log($\sec^2$ x)+c

7. $\int e^{3-4x}dx$ is _____

a) $3e^{3-4x}$ +c
b) $-4e^{3-4x}$ +c
c) $\frac{e^{3-4x}}{4}+c$ d) $\frac{e^{3-4x}}{-4}+c$

8. The value of $\int \sec^2(x-5)dx$ is

a) tan(x-5)+c b) sec(x-5).tan(x-5)+c
c) $\tan^2$(x-5)+c d) none of these

9. The value of $\int \frac{1}{10x+3}dx$ is

a) $\frac{1}{10x}+c$ b) $\frac{-1}{(10x+3)^2}+c$ c) $\frac{3}{10x+3}+c$ d) $\frac{\log(10x+3)}{10}+c$

10.Integral of $\frac{9\cos x}{5\sin^2 x}$ with respect to x is

a) $\frac{-9}{5}\cos ecx+c$ b) $\frac{9}{5}\cos ecx+c$ c) $\frac{-9\sin x}{5\cos^2 x}+c$ d) $\frac{9\sin x}{5\cos^2 x}+c$

11.The value of $\int \frac{2x}{1+x^2}dx$ is

a) $1+x^2+c$ b) 2x+c c) log($1+x^2$)+c d) log 2x+c

12. $\int \frac{e^x}{e^x+1}dx$ is equal to

a) log(e^x+1)+c b) log e^x + c c) e^x + c d) e^x +1 + c

13.The value of $\int \frac{1}{x^2}dx$ is

a) $\frac{1}{x}+c$ b) $\frac{2}{x}+c$ c) $\frac{-2}{x^2}+c$ d) $\frac{-1}{x}+c$

14.The integral of $4\cos ec^2 x$ with respect to x is

a) $\sec^2$ x +c b) 4 $\sec^2$ x +c c) – cot x +c d) – 4 cot x +c

15.Find the integral of $\frac{9}{\sin^2 x}$ w.r.t x.

16.Evaluate $\int(4\sec^2 x)dx$

17. Evaluate $\int \frac{3}{x} dx$

18. Find $\int \frac{1}{5e^{-x}} dx$

19. Find $\int \left(7^x - 10.9^x\right) dx$

20. Evaluate $\int \left(\tan^2 x\right) dx$

<u>Two Marks Questions:</u>

Evaluate the following

1. $\int \frac{x^5 + 5x^2 - 7x}{\sqrt{x}} dx$

2. $\int \left(x + \frac{1}{x}\right)^2 dx$

3. $\int \left(x - \frac{1}{x}\right)^3 dx$

4. $\int \left(\sqrt{1 - \sin 2x}\right) dx$

5. $\int \left(\sqrt{1 - \cos 2x}\right) dx$

6. $\int \left(\sqrt{1 + \sin 2x}\right) dx$

7. $\int \sec x \left(\sec x - \tan x\right) dx$

8. $\int \frac{\cos^2 x}{1 + \sin x} dx$

9. $\int \sqrt{x} \left(x - \frac{1}{x}\right) dx$

10. $\int \left(\sin^2 x\right) dx$

11. $\int \left(\sin^3 x\right) dx$

12. $\int \left(\cos^2 x\right) dx$

13. $\int \left(\cos^3 x\right) dx$

14. $\int \left(\tan^2 x\right) dx$

15. $\int \tan^2 \left(3 - 2x\right) dx$

16. $\int \operatorname{cosec}\left(3 - 5x\right).\cot\left(3 - 5x\right) dx$

17. $\int \frac{3x + 2}{3x^2 + 4x - 5} dx$

18. $\int \frac{4x + 3}{2x^2 + 3x + 5} dx$

19. $\int \frac{e^x - 1}{e^x - x} dx$

20. $\int \frac{1}{x\left(2\log x + 5\right)} dx$

21. $\int \frac{1 + \tan x}{1 - \tan x} dx$

22. $\int \frac{\tan x}{2 + \log\left(\sec x\right)} dx$

23. $\int e^x \left(\sqrt{e^x + 1}\right) dx$

24. $\int \frac{1}{x} \left(\log x\right)^3 dx$

25. $\int \left(2x + 3\right)\left(x^2 + 3x + 5\right)^{3/2} dx$

26. $\int \left(\log x\right) dx$

Three Marks Questions:

Evaluate the following

1. $\int (x.\sin x)dx$
2. $\int (x^2.\log x)dx$
3. $\int (x.\log x)dx$
4. $\int (x.e^x)dx$
5. $\int (x^2.\cos x)dx$
6. $\int (x.e^{2x})dx$
7. $\int (x.\sec^2 x)dx$
8. $\int \frac{4x+5}{(x-1)(x+2)}dx$
9. $\int \frac{3x+2}{(2x+3)(3x-1)}dx$
10. $\int \frac{3}{x^2+5x+6}dx$
11. $\int \frac{3}{(x+1)(x+2)}dx$
12. $\int \frac{x+2}{(2x-1)(x-3)}dx$
13. $\int \frac{1}{x(x+1)(x+2)}dx$
14. $\int (6x+5)(3x^2+5x-4)^{5/3}dx$
15. $\int \operatorname{cosec}^2 x\left(\sqrt{1+\cot x}\right)dx$
16. $\int \frac{1+e^x}{(x+e^x)^5}dx$
17. $\int \frac{2}{x}(\log x)^2 dx$
18. $\int \cos^2 x(\sin x)dx$
19. $\int x\left(\sqrt{3x^2+5}\right)dx$
20. $\int \frac{1}{\sqrt{x}+x}dx$
21. $\int \frac{1}{x(2\log x+5)}dx$
22. $\int \frac{3x^2}{1+x^2}dx$
23. $\int \frac{2x+5}{x^2+5x+3}dx$
24. $\int (\cos 7x.\cos 6x)dx$
25. $\int (\sin 2x.\cos 3x)dx$
26. $\int \frac{2x}{2x+3}dx$
27. $\int \frac{2x+5}{3x+4}dx$
28. $\int \left(\sqrt{1+\sin 2x}\right)dx$
29. $\int \frac{x}{\sqrt{x-5}}dx$
30. $\int \cot^2(5x+3)dx$

Chapter 21: DEFINITE INTEGRALS AND ITS APPLICATIONS TO AREAS

One Mark Questions:

Evaluate the following

1. $\int_0^1 x^2 dx$
2. $\int_0^{\pi/2} (\sin x) dx$
3. $\int_1^2 (x) dx$
4. $\int_0^1 e^{2x} dx$
5. $\int_0^1 e^x dx$
6. $\int_1^2 7x^2 dx$
7. $\int_0^{\pi/2} \cos x dx$
8. $\int_0^{\pi/2} \sin x dx$
9. $\int_1^2 \frac{1}{2x+3} dx$

Two Marks Questions:

1. Evaluate $\int_{-\pi/4}^{\pi/4} (\sec^2 x) dx$
2. Evaluate $\int_1^2 (x + e^x) dx$
3. Find $\int_0^{\pi/2} (\sin x + \cos x) dx$
4. Find the value of $\int_0^1 \frac{e^x + 1}{e^x} dx$
5. Evaluate $\int_1^2 x.e^x dx$
6. Evaluate $\int_1^2 (\log x) dx$
7. Evaluate $\int_0^{\pi/2} (\sin 2x) dx$
8. Evaluate $\int_0^4 (\sqrt{x} - x) dx$
9. Find the area bounded by x-axis, the curve $y = x^2$ and ordinates $x = 0$ and $x = 1$.
10. Find the area bounded by the curve $x^2 = 8y$, y-axis and the lines $y = 3$ and $y = 6$.
11. Find the area enclosed by the curve $y = x^2$, x-axis and the lines $x = 1$ and $x = 3$.
12. Find the area enclosed by the curve $x = 2y^2$, y-axis and the lines $y = 2$ and $y = 4$.

Three Marks Questions:

Evaluate the following

1. $\int_0^1 (6x+1)\left(\sqrt{3x^2+x+5}\right)dx$
2. $\int_{-1}^1 (10x+3)\left(\sqrt{5x^2+3x+7}\right)dx$
3. $\int_0^{\pi/2} x(\sin x)\,dx$
4. $\int_0^{\pi/2} x(\cos x)\,dx$
5. $\int_1^2 \frac{2x+5}{x^2+5x+3}dx$
6. $\int_0^3 \frac{x+3}{x+2}dx$
7. $\int_0^1 \frac{2x+3}{3x+5}dx$
8. $\int_2^3 \frac{1}{(x+1)(x+2)}dx$

Two and Three Marks Questions:

1. If the MC of a firm is $f'(x)=10+6x-6x^2$ $(x=output)$, find the TC given fixed cost is ₹125.
2. If the MC of a product is $2x^2-4x$, find TC of producing 16 units, x is the output.
3. If MC function is $3x^2-x+5$, x=output, then find TC, AC, TVC and average variable cost, given that its fixed cost is Rs.25.
4. If MR is $f'(x)=20-\frac{x}{20}$, find the TR and AR obtained from an output of 30 units.
5. If $MR=10x-x^2$, find the TR. Also find its demand function.

Five Marks Questions:

1. Find the area bounded by the parabola $y^2=4ax$ and its latus rectum.
2. Find the area between the curves $y^2=4x$ and $x^2=4y$.
3. Find the area bounded by the curve $y^2=5x$ and the line $y=x$.
4. Find the area enclosed by the parabola $y^2=4x$ and the line $x-y=0$.
5. Find the area enclosed between the parabolas $y^2=4ax$ and $x^2=4ay$.